

Handover — Sport Analytics Tool, Ben Swartz Paste this at the start of a new conversation to resume work. --- ## Project COM S3011A Software Design...

PASTED

Synthesized handover context and prioritized next concrete step

Context loaded. Picking up #27 with C3, C4, D2, the nav check, then the merge — in that order, since C3/C4 may force migration changes on the same branch, and D2's per-delivery storage figure will be more honest measured after three fixtures than after one.

Two things before we start, both flags rather than actions:

The **IT20 misclassification is worse than a downloader bug**. If Nadav's downloader has been using `match_type: "IT20"` as the international filter, then whatever corpus is currently in `data/` may be the wrong corpus, and any count derived from it — including your 13,953 and 3,193,996 — is suspect. Worth confirming which filter produced the local corpus before D2 uses it for a storage projection.

Password reset and account deletion are named key requirements in the project brief, not nice-to-haves, and neither the Supabase Auth ruling nor ADR-004 status is resolved with 18 days to Sprint 1. Not yours to fix, but it belongs on Tuesday's stakeholder agenda with a named owner, not in a handover list with no date.

Step 1 — find a super-over fixture and confirm what shape it is.

In Cricsheet, a super over appears as an additional innings object carrying `super_over: true`. Before ingesting anything, I want to see the structure you're actually feeding the parser, because your innings ordinal and `(fixture, innings ordinal, over, position)` identity have to accommodate a third and fourth innings on a T20 fixture.

From the repository root, run something that lists a few candidates and dumps the innings-level keys of one of them — team, `super_over` flag, and over count per innings. A short Python script against `data/` is fine; PowerShell will struggle with the JSON nesting.

Post me the file ID you pick and the innings-level structure it prints. I'll tell you what to check next.

wait, tell me what to do step by step

Step 1 — build the list of super-over fixtures and look at one.

Create this file at `scripts/find_super_overs.py`:

```
python

import json, sys
from pathlib import Path

DATA = Path("data")
matches = sorted(DATA.rglob("*.json"))
print(f"scanned root: {DATA.resolve()}")
print(f"files found: {len(matches)}")

hits = []
for p in matches:
    try:
        with p.open(encoding="utf-8") as f:
            d = json.load(f)
    except Exception as e:
        print(f"UNREADABLE {p.name}: {e}")
        continue
    innings = d.get("innings", [])
    if any(i.get("super_over") for i in innings):
        hits.append((p, d))

print(f"super-over matches: {len(hits)}")
for p, _ in hits[:10]:
    print(" ", p.name)

if not hits:
    sys.exit(0)

p, d = hits[0]
print(f"\n--- structure of {p.name} ---")
info = d.get("info", {})
print("match_type:", info.get("match_type"), "| team_type:", info.get("team_type"))
print("teams:", info.get("teams"))
print("outcome:", info.get("outcome"))
for idx, i in enumerate(d.get("innings", []), start=1):
    overs = i.get("overs", [])
    keys = sorted(k for k in i.keys() if k not in ("overs",))
    print(f"innings {idx}: team={i.get('team')!r} super_over={i.get('super_over')} "
          f"overs={len(overs)} deliveries={sum(len(o.get('deliveries', [])) for o in overs)} "
          f"other_keys={keys}")
    for o in overs:
        print(f"    over={o.get('over')!r} ({type(o.get('over')).__name__}) "
              f"deliveries={len(o.get('deliveries', []))}")
```

Run it from the repository root:

```
powershell  
  
python scripts/find_super_overs.py
```

Paste me the whole output. Three things I want to read in it: whether the count comes to 99 as your handover says, whether the super over is a separate innings object or folded into an existing one, and whether the super-over innings restarts its over numbering at 0 — because if it does, (fixture, innings ordinal, over, position) stays sound only if the super over gets its own ordinal rather than being appended to innings 1 and 2.

Don't commit the file yet. If it earns a place it belongs alongside db:validate as ingestion evidence, and that's a separate decision from #27. When it is committed it'll need Assisted-by: Claude-Web[Claude Opus 5] in the commit body.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> python scripts/find_super_overs.py  
scanned root: C:\Users\BenSwartz\Sport-Analytics-Tool\data  
files found: 13955  
Traceback (most recent call last):  
  File "C:\Users\BenSwartz\Sport-Analytics-Tool\scripts\find_super_overs.py", line  
17, in <module>  
    innings = d.get("innings", [])  
            ^^^^^  
AttributeError: 'list' object has no attribute 'get'  
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Identified JSON data structure mismatch in script

That's informative rather than annoying. You have 13,955 .json files but the handover says 13,953 matches — so at least one, probably two, of those files isn't a match. The script hit one and died. Fix is to skip non-objects and name them.

Replace the loop body so it tolerates and reports them. Change these lines:

```
python  
  
except Exception as e:  
    print(f"UNREADABLE {p.name}: {e}")  
    continue  
innings = d.get("innings", [])
```

to:

```
python
```

```

except Exception as e:
    print(f"UNREADABLE {p.name}: {e}")
    continue
if not isinstance(d, dict):
    print(f"NOT-A-MATCH {p.relative_to(DATA)}: top level is {type(d).__name__}, "
          f"len={len(d)}")
    continue
innings = d.get("innings", [])

```

Then run it again:

```

powershell

python scripts/find_super_overs.py

```

Paste the output. I want the **NOT-A-MATCH** lines specifically as well as the super-over results — if those two files are a downloader manifest or an index, fine, but if they're something the ingestion path would try to read as a match, that's a defect in Nadav's downloader worth an issue.

```

PS C:\Users\BenSwartz\Sport-Analytics-Tool> python scripts/find_super_overs.py
scanned root: C:\Users\BenSwartz\Sport-Analytics-Tool\data
files found: 13955
NOT-A-MATCH cricsheet\manifest.json: top level is list, len=13953
super-over matches: 99
423788.json
1183544.json
1345425.json
1169389.json
1208096.json
1516546.json
1023647.json
1195586.json
654069.json
756767.json
--- structure of 423788.json ---
match_type: T20 | team_type: international
teams: ['New Zealand', 'Australia']
outcome: {'eliminator': 'New Zealand', 'result': 'tie'}
innings 1: team='New Zealand' super_over=None overs=20 deliveries=125
other_keys=['powerplays', 'team']
over=0 (int) deliveries=7
over=1 (int) deliveries=6
over=2 (int) deliveries=6
over=3 (int) deliveries=6
over=4 (int) deliveries=6
over=5 (int) deliveries=6
over=6 (int) deliveries=6
over=7 (int) deliveries=6

```

```

over=8 (int) deliveries=6
over=9 (int) deliveries=6
over=10 (int) deliveries=6
over=11 (int) deliveries=6
over=12 (int) deliveries=6
over=13 (int) deliveries=6
over=14 (int) deliveries=7
over=15 (int) deliveries=6
over=16 (int) deliveries=6
over=17 (int) deliveries=6
over=18 (int) deliveries=9
over=19 (int) deliveries=6
innings 2: team='Australia' super_over=None overs=20 deliveries=125 other_keys=
['powerplays', 'target', 'team']
over=0 (int) deliveries=6
over=1 (int) deliveries=7
over=2 (int) deliveries=6
over=3 (int) deliveries=6
over=4 (int) deliveries=6
over=5 (int) deliveries=6
over=6 (int) deliveries=6
over=7 (int) deliveries=7
over=8 (int) deliveries=6
over=9 (int) deliveries=7
over=10 (int) deliveries=6
over=11 (int) deliveries=7
over=12 (int) deliveries=6
over=13 (int) deliveries=6
over=14 (int) deliveries=6
over=15 (int) deliveries=6
over=16 (int) deliveries=6
over=17 (int) deliveries=6
over=18 (int) deliveries=7
over=19 (int) deliveries=6
innings 3: team='Australia' super_over=True overs=1 deliveries=6 other_keys=
['super_over', 'team']
over=0 (int) deliveries=6
innings 4: team='New Zealand' super_over=True overs=1 deliveries=5 other_keys=
['super_over', 'team']
over=0 (int) deliveries=5
PS C:\Users\BenSwartz\Sport-Analytics-Tool> python scripts/find_super_overs.py

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```

Identified schema gaps and data structure inconsistencies in cricket analytics

Good result — and it changes what #27 needs before it can merge.

What the output establishes:

1. `manifest.json` is in the corpus directory and is a list of 13,953 entries. If `db:ingest` globs `data/**/*.json` it will hit this file exactly as your script did. Not a blocker for C3, but it's a real defect and I want to check it.

2. **99 super-over matches** — matches the handover figure, so the corpus is what you thought on that count at least.
3. **The super over is separate innings objects**, ordinals 3 and 4, `super_over: true`, over numbering restarting at `0`. Your identity `(fixture, innings ordinal, over, position)` holds without change, because the innings ordinal disambiguates over 0.
4. **The batting order reverses**. New Zealand batted first in the match; Australia bats first in the super over. Anything inferring team from innings parity is wrong.
5. **The outcome shape is different**: `result: 'tie'` with an `eliminator` key, no `winner`.

The thing that concerns me most is not ingestion, it's derivation. Super-over runs and wickets do not count toward batting or bowling statistics in cricket. If the innings table has no `super_over` flag, then a match ingests cleanly, C3 passes, and every career aggregate downstream is quietly wrong by six balls per super over across 99 matches. That's exactly the class of error the project brief calls out — a derived statistic is only worth anything if it is correct.

So before you ingest anything, we check whether the schema can represent this.

Step 2 — inspect what the applied migration actually stores for innings and outcome.

From the repository root:

```
powershell
```

```
Select-String -Path database\migrations\*.sql,database\migrations\*.js,database\migrat
```

If your migrations are `node-pg-migrate` plain SQL they'll be `.sql`; the command covers all three so it doesn't matter.

Paste the output. Specifically I'm looking for: does the `innings` table have a `super_over` boolean, and does the fixture table have somewhere to put `eliminator` and a `result` of `tie`. If either is missing, we amend the migration on this branch rather than merging and fixing later — the migration hasn't left your branch, so amending it is cheaper now than a follow-up migration after merge.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Select-String -Path database\migrations\*.sql,database\migrations\*.js,database\migrations\*.ts -Pattern...
```

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Validated schema integrity and identified minor data constraints

The schema already accommodates all of it. Nothing to amend:

- `innings.is_super_over` exists, and the column comment already anticipates four innings.

- `outcome_kind` includes `'tie'`, and `eliminator_id` exists on `fixture`.
- `fixture_winner_ck` reads `(outcome = 'won') = (winner_id IS NOT NULL)` — a tie with a null winner and a set eliminator satisfies it.
- `innings_miscounted_over` is already there for C4.

So C3 is a genuine confirmation run rather than a redesign. Two notes I'm parking, not acting on:

No constraint stops `eliminator_id` being set when `outcome = 'won'`. Looseness, not a defect, and the migration is still on your branch so it's cheap to tighten. I'd rather finish C3 and C4 first and decide then, because C4 may surface a second constraint worth adding in the same edit.

`is_super_over` solves storage, not derivation. Every batting and bowling aggregate must exclude super-over innings, and `delivery_current` as a view over deliveries won't do that on its own. Not #27's scope, but it needs to be an issue against whoever owns derivation before it silently corrupts career figures across 99 matches.

Step 3 — ingest the super-over match.

powershell

```
npm run db:ingest --workspace=@sport-analytics/backend -- data/cricsheet/423788.json
```

Adjust the path if `423788.json` sits somewhere other than `data/cricsheet/` — your script printed bare filenames, so confirm with `ls data -Recurse -Filter 423788.json` first if you're unsure. If `db:ingest` takes its argument differently, paste me its definition from `package.json` rather than guessing.

Paste the full output including any error. What I expect to see is four innings created, 261 deliveries (125 + 125 + 6 + 5), and the fixture row carrying `tie` with New Zealand as eliminator. What I'm watching for is the ingestion assuming two innings, or inferring the batting side from innings parity — the super over reverses the order, so a parity assumption puts Australia's super over under New Zealand.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:ingest --
workspace=@sport-analytics/backend -- data/cricsheet/423788.json
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts data/cricsheet/423788.json
node:fs:556
```

```
  return binding.open(
    ^
```

```
Error: ENOENT: no such file or directory, open 'C:\Users\BenSwartz\Sport-
Analytics-Tool\apps\backend\data\cricsheet\423788.json'
  at Object.openSync (node:fs:556:18)
  at readFileSync (node:fs:440:35)
  at <anonymous> (C:\Users\BenSwartz\Sport-Analytics-
Tool\apps\backend\scripts\ingest-match.ts:51:13)
```

```
at Object.<anonymous> (C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend\scripts\ingest-match.ts:432:4)
at Module._compile (node:internal/modules/cjs/loader:1829:14)
at Object.transformer (C:\Users\BenSwartz\Sport-Analytics-Tool\node_modules\tsx\dist\register-BcxQ17Jo.cjs:9:3339)
at Module.load (node:internal/modules/cjs/loader:1552:32)
at Module._load (node:internal/modules/cjs/loader:1354:12)
at wrapModuleLoad (node:internal/modules/cjs/loader:255:19)
at loadCJSModuleWithModuleLoad
(node:internal/modules/esm/translators:327:3) {
  errno: -4058,
  code: 'ENOENT',
  syscall: 'open',
  path: 'C:\\Users\\BenSwartz\\Sport-Analytics-Tool\\apps\\backend\\data\\cricsheet\\423788.json'
}
Node.js v25.9.0
npm error Lifecycle script db:ingest failed with error:
npm error code 1
npm error path C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend
npm error workspace @sport-analytics/backend@0.1.0
npm error location C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend
npm error command failed
npm error command C:\windows\system32\cmd.exe /d /s /c tsx --env-file=.env scripts/ingest-match.ts data/cricsheet/423788.json
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

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Diagnosed path resolution issue stemming from working directory mismatch

`npm run --workspace` sets the working directory to `apps/backend`, so your repository-root-relative path resolved against the wrong base. The file wasn't missing.

Step 3a — confirm where the file actually is:

powershell

```
ls data -Recurse -Filter 423788.json | Select-Object FullName
```

Step 3b — ingest using an absolute path, which sidesteps the cwd question entirely:

powershell

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

Substitute the `FullName` that 3a prints if it differs.

Paste both outputs.

Two flags, neither blocking:

`db:ingest` resolving relative paths against `apps/backend` is a trap for anyone following the documented commands from the repository root, which is how your handover describes them. Worth having the script resolve against the repository root, or worth a line in the docs. Small, and it's yours — fold it into #27 or raise it separately, your call.

You're on Node 25.9.0 and the project targets Node 20. Divergence between your local runtime and whatever Azure and CI pin is the kind of thing that surfaces as an unreproducible failure in week eight. Not part of #27 and not urgent today, but it should be an issue with an owner.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> ls data -Recurse -Filter 423788.json |
Select-Object FullName
FullName
-----
C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\australia-in-ne...
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

PowerShell truncated the column. Get the full value:

```
powershell

ls data -Recurse -Filter 423788.json | Select-Object -ExpandProperty FullName
```

Paste that, then ingest with it:

```
powershell

npm run db:ingest --workspace=@sport-analytics/backend -- "<paste the full path here>"
```

Worth noting from the fragment you already have: the corpus is organised into per-series subdirectories under `data/cricsheet/matches/`, not a flat directory. That's fine, but it means anything reading the corpus must recurse — your `find_super_overs.py` did, `db:ingest` takes one file so it doesn't care, but a batch import path for Sprint 2 will. And `manifest.json` sitting a level up from `matches/` is likely deliberate on the downloader's part rather than the defect I flagged earlier, since a recursive glob rooted at `matches/` would never reach it. I'd downgrade that to "check what `db:ingest`'s eventual batch equivalent globs" rather than "Nadav's downloader is broken."

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> ls data -Recurse -Filter 423788.json |
Select-Object -ExpandProperty FullName
C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\australia-in-new-
zealand-t20i-series\423788.json
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Step 3b — ingest it:

```
powershell
```

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

Paste the full output, error or not.

What I'll be checking against: four innings rows, 261 deliveries total (125 + 125 + 6 + 5), innings 3 batting Australia with `is_super_over = true`, innings 4 batting New Zealand, fixture outcome `tie` with `winner_id` null and `eliminator_id` set to New Zealand. Plus the shared database moving from 237 deliveries to 498, which matters for anyone else counting.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:ingest --
workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\australia-in-new-zealand-t20i-series\423788.json"
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\australia-in-new-zealand-t20i-series\423788.json
Ingested 423788
  fixture_id    : 3
  people resolved : 27
  deliveries added : 261
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Investigated fixture sequencing discrepancy and verified delivery ingestion accuracy

261 is right, and person resolution at 27 is plausible for two XIs plus substitutes.

Note `fixture_id : 3`, not 2 — the identity sequence has advanced past a fixture that isn't there, probably from a rolled-back or failed earlier attempt. Harmless in itself, but it means fixture IDs are not a count of fixtures, and nobody should write a check that assumes they are.

Now verify by SQL rather than by trusting the ingest summary — the same standard you held 729307 to. You'll need ad-hoc SQL again for C4 and for D2's `pg_total_relation_size` measurements, so it's worth a small reusable script rather than three one-offs.

Step 4 — create `apps/backend/scripts/sql.ts` :

```
ts
```

```

import { readFileSync } from "node:fs";
import { Client } from "pg";

const arg = process.argv[2];
if (!arg) {
  console.error("usage: db:sql \"<query>\" | db:sql --file <path>");
  process.exit(1);
}

const sql =
  arg === "--file" ? readFileSync(process.argv[3], "utf8") : arg;

const client = new Client({
  connectionString: process.env.DATABASE_URL,
  ssl: {
    ca: readFileSync(new URL("../certs/supabase-ca.crt", import.meta.url), "utf8"),
    rejectUnauthorized: true,
  },
});

await client.connect();
try {
  const result = await client.query(sql);
  const results = Array.isArray(result) ? result : [result];
  for (const r of results) {
    if (r.rows?.length) console.table(r.rows);
    else console.log(`${r.command}: ${r.rowCount ?? 0}`);
  }
} finally {
  await client.end();
}

```

Then add to `apps/backend/package.json` scripts:

```

json
"db:sql": "tsx --env-file=.env scripts/sql.ts"

```

Two things to check rather than assume: that your existing scripts construct the TLS config this way — open `scripts/ingest-match.ts` and copy its client setup verbatim if it differs, since it's the one that's known to work against the pooler. And that `pg` is a direct dependency of the backend workspace, not just transitive.

Tell me if the client setup in `ingest-match.ts` differs from the above and I'll adjust before you run anything.

I need a bit more than "done" here — did the client setup in `ingest-match.ts` match, or did you have to change it? If you changed it, paste what you used. It matters because the next several steps all run through this script and I'd rather know its actual shape than assume.

Assuming it works, **Step 5 — verify the super-over ingestion:**

```
powershell
```

```
npm run db:sql --workspace=@sport-analytics/backend -- "SELECT i.ordinal, t.name AS ba
```

Two guesses in that query I can't confirm from what I've seen: that `team` has a `name` column, and that the delivery-liveness column is `superseded_at` on `delivery` (your migration output showed `WHERE superseded_at IS NULL` on the indexes, so that one is solid). If `team.name` is wrong, the error will tell you the real column and you can substitute.

Paste the table. I want four rows: ordinals 0 and 1 at 125 each with `is_super_over` false, then Australia at 6 and New Zealand at 5, both true. If the super-over batting teams come out the other way round, the ingestion is inferring team from parity and that's a defect to fix before merge.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --
workspace=@sport-analytics/backend -- "SELECT i.ordinal, t.name AS
batting_team, i.is_super_over, count(d.delivery_id) AS deliveries FROM innings i
JOIN team t ON t.team_id = i.batting_team_id LEFT JOIN delivery d ON
d.innings_id = i.innings_id AND d.superseded_at IS NULL WHERE i.fixture_id = 3
GROUP BY i.ordinal, t.name, i.is_super_over ORDER BY i.ordinal"
> @sport-analytics/backend@0.1.0 db:sql
> tsx --env-file=.env scripts/sql.ts SELECT i.ordinal, t.name AS batting_team,
i.is_super_over, count(d.delivery_id) AS deliveries FROM innings i JOIN team t ON
t.team_id = i.batting_team_id LEFT JOIN delivery d ON d.innings_id = i.innings_id
AND d.superseded_at IS NULL WHERE i.fixture_id = 3 GROUP BY i.ordinal, t.name,
i.is_super_over ORDER BY i.ordinal
node:internal/modules/run_main:107
  triggerUncaughtException(
    ^
Error: Transform failed with 3 errors:
C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend\scripts\sql.ts:21:0:
ERROR: Top-level await is currently not supported with the "cjs" output format
C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend\scripts\sql.ts:23:17:
ERROR: Top-level await is currently not supported with the "cjs" output format
C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend\scripts\sql.ts:30:2:
ERROR: Top-level await is currently not supported with the "cjs" output format
    at failureErrorWithLog (C:\Users\BenSwartz\Sport-Analytics-
Tool\node_modules\esbuild\lib\main.js:1748:15)
    at C:\Users\BenSwartz\Sport-Analytics-
Tool\node_modules\esbuild\lib\main.js:1017:50
    at responseCallbacks.<computed> (C:\Users\BenSwartz\Sport-Analytics-
```

```
Tool\node_modules\esbuild\lib\main.js:884:9)
  at handleIncomingPacket (C:\Users\BenSwartz\Sport-Analytics-
Tool\node_modules\esbuild\lib\main.js:939:12)
  at Socket.readFromStdout (C:\Users\BenSwartz\Sport-Analytics-
Tool\node_modules\esbuild\lib\main.js:862:7)
  at Socket.emit (node:events:509:20)
  at addChunk (node:internal/streams/readable:564:12)
  at readableAddChunkPushByteMode (node:internal/streams/readable:515:3)
  at Readable.push (node:internal/streams/readable:395:5)
  at Pipe.onStreamRead (node:internal/stream_base_commons:189:23) {
  name: 'TransformError'
}
Node.js v25.9.0
npm error Lifecycle script db:sql failed with error:
npm error code 1
npm error path C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend
npm error workspace @sport-analytics/backend@0.1.0
npm error location C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend
npm error command failed
npm error command C:\windows\system32\cmd.exe /d /s /c tsx --env-file=.env
scripts/sql.ts ^^^"SELECT^^^ i.ordinal,^^^ t.name^^^ AS^^^ batting_team,^^^
i.is_super_over,^^^ count^^^(d.delivery_id^^^)^^^ AS^^^ deliveries^^^
FROM^^^ innings^^^ i^^^ JOIN^^^ team^^^ t^^^ ON^^^ t.team_id^^^ =^^^
i.batting_team_id^^^ LEFT^^^ JOIN^^^ delivery^^^ d^^^ ON^^^
d.innings_id^^^ =^^^ i.innings_id^^^ AND^^^ d.superseded_at^^^ IS^^^
NULL^^^ WHERE^^^ i.fixture_id^^^ =^^^ 3^^^ GROUP^^^ BY^^^ i.ordinal,^^^
t.name,^^^ i.is_super_over^^^ ORDER^^^ BY^^^ i.ordinal^^^"
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

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Two problems, and the second one is the reason I asked what `ingest-match.ts` looks like.

Problem 1: CJS. The workspace resolves as CommonJS, so top-level `await` fails, and `import.meta.url` won't work either. `ingest-match.ts` evidently avoids both — which is why it runs.

Problem 2: PowerShell mangled the argument. Look at the command npm actually ran: `^^^"SELECT^^^ i.ordinal,...`. The quoting is destroyed. Passing SQL as a command-line argument through `npm run` on Windows is going to keep biting us, and we have several more queries to run. Use a file instead.

Step 4a — replace `apps/backend/scripts/sql.ts` entirely with this:

```
ts
```

```

import { readFileSync } from "node:fs";
import { resolve } from "node:path";
import { Client } from "pg";

async function main() {
  const path = process.argv[2];
  if (!path) {
    console.error("usage: npm run db:sql -- <path-to-.sql-file>");
    process.exit(1);
  }

  const sql = readFileSync(resolve(process.cwd(), path), "utf8");

  const client = new Client({
    connectionString: process.env.DATABASE_URL,
    ssl: {
      ca: readFileSync(resolve(__dirname, "../certs/supabase-ca.crt"), "utf8"),
      rejectUnauthorized: true,
    },
  });

  await client.connect();
  try {
    const result = await client.query(sql);
    const results = Array.isArray(result) ? result : [result];
    for (const r of results) {
      if (r.rows?.length) console.table(r.rows);
      else console.log(`${r.command}: ${r.rowCount ?? 0}`);
    }
  } finally {
    await client.end();
  }
}

main().catch((err) => {
  console.error(err);
  process.exit(1);
});

```

Before you run it: open `apps/backend/scripts/ingest-match.ts` and paste me its `new Client({...})` block. I'm reconstructing the TLS and pooler configuration from your handover notes rather than from the code, and that connection is the one thing in this project that's documented as fragile. If it differs from what I've written, mine is wrong, not yours.

Note the script now takes a file path relative to `apps/backend` — the same cwd trap as `db:ingest`. I'll give absolute paths in the queries that follow.

```
/** * Load a single Cricshet match into the delivery event schema. * * This script exists to validate the schema against a published scorecard. It is * not t...
```

PASTED

Reconciled module patterns and identified data handling inconsistencies

Your ingest script settles it: `import.meta.url` works fine under tsx here — only top-level `await` breaks. So the fix is just to wrap in a `main()`, same as `ingest-match.ts` does, and keep the TLS block identical to yours.

Step 4a — replace `apps/backend/scripts/sql.ts` with this:

ts

```

/**
 * Run an ad-hoc SQL file against the project database.
 *
 * Exists to support schema validation and measurement. It is not part of the
 * application and holds no business logic.
 *
 * Usage:
 *   npm run db:sql --workspace=@sport-analytics/backend -- <path-to-query.sql>
 *
 * The path is resolved relative to apps/backend.
 */

import { readFileSync } from 'node:fs';
import { resolve } from 'node:path';
import { Client } from 'pg';

const queryPath = process.argv[2];
if (!queryPath) {
  console.error('Usage: db:sql -- <path-to-query.sql>');
  process.exit(1);
}

const client = new Client({
  connectionString: process.env.DATABASE_URL,
  ssl: {
    rejectUnauthorized: true,
    ca: readFileSync(new URL('../certs/supabase-ca.crt', import.meta.url), 'utf8'),
  },
});

async function main(): Promise<void> {
  const sql = readFileSync(resolve(process.cwd(), queryPath), 'utf8');
  await client.connect();
  const result = await client.query(sql);
  for (const part of Array.isArray(result) ? result : [result]) {
    if (part.rows?.length) console.table(part.rows);
    else console.log(`${part.command}: ${part.rowCount ?? 0}`);
  }
}

main()
  .catch((error: unknown) => {
    console.error(error instanceof Error ? error.message : error);
    process.exitCode = 1;
  })
  .finally(() => {

```

```
void client.end();
});
```

Keep the `"db:sql": "tsx --env-file=.env scripts/sql.ts"` line you added.

Step 4b — create `apps/backend/scripts/queries/c3-super-over.sql`:

```
sql

SELECT i.ordinal,
       t.name AS batting_team,
       i.is_super_over,
       count(d.delivery_id) AS deliveries
FROM innings i
JOIN team t ON t.team_id = i.batting_team_id
LEFT JOIN delivery d
      ON d.innings_id = i.innings_id
      AND d.superseded_at IS NULL
WHERE i.fixture_id = 3
GROUP BY i.ordinal, t.name, i.is_super_over
ORDER BY i.ordinal;
```

Step 4c — run it:

```
powershell

npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/c3-super-over.s
```

Paste the table.

Two things I noticed reading the script, neither blocking C3:

ball_number carries no information. The fallback is ``${over.over}.${position + 1}``, and Cricsheet has no per-delivery ball label, so **actual_delivery** is presumably always undefined. That means **ball_number** is a string restatement of **over_number** and **position_in_over** — it never repeats, so the 20.4% repetition finding that motivated your natural key cannot be reproduced from the database. Either the column should hold the computed legal-ball number (which does repeat, and is what a scorecard prints), or it should be dropped. Worth resolving before merge since it's a documented design justification.

The miscounted_overs key may be off by one. The script does `Number(over)` on the object key and stores it as **over_number**, which elsewhere is zero-based from `over.over`. If Cricsheet keys that map by the displayed (one-based) over, every miscounted-over row points at the wrong over. C4 is exactly the run that will show this — I'll have you check it explicitly rather than trusting the insert.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --
workspace=@sport-analytics/backend -- scripts/queries/c3-super-over.sql
> @sport-analytics/backend@0.1.0 db:sql
> tsx --env-file=.env scripts/sql.ts scripts/queries/c3-super-over.sql
```

(index)	ordinal	batting_team	is_super_over	deliveries
0	0	'New Zealand'	false	'125'
1	1	'Australia'	false	'125'
2	2	'Australia'	true	'6'
3	3	'New Zealand'	true	'5'

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Show more

Exactly right, including the reversal — Australia bats the super over having bowled the second innings. Nothing is inferring team from parity. **C3 passes on ingestion.**

One gap before you write it up. C3's claim is that a super over ingests correctly, and correctness includes the fixture-level outcome, which is where the tie-plus-eliminator shape is unusual and where `fixture_winner_ck` gets its only real exercise. Cheap to check now.

Step 5 — create `apps/backend/scripts/queries/c3-outcome.sql`:

```
sql

SELECT f.outcome,
       w.name AS winner,
       e.name AS eliminator,
       f.outcome_by_runs,
       f.outcome_by_wickets,
       f.decided_by_bowl_out
FROM fixture f
LEFT JOIN team w ON w.team_id = f.winner_id
LEFT JOIN team e ON e.team_id = f.eliminator_id
WHERE f.fixture_id = 3;
```

Run it:

```
powershell

npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/c3-outcome.sql
```

Expected: `tie`, winner null, eliminator New Zealand, both margins null.

If that comes back clean, C3 is done and we move to C4 — which is the more interesting one, because of the `miscounted_overs` key indexing I flagged and because the source types the

ball count inconsistently.

Note for when you commit these: `scripts/queries/` is new, the three files are AI-produced, and the commit body needs `Refs #27` plus `Assisted-by: Claude-Web[Claude Opus 5]`.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --
workspace=@sport-analytics/backend -- scripts/queries/c3-outcome.sql
> @sport-analytics/backend@0.1.0 db:sql
> tsx --env-file=.env scripts/sql.ts scripts/queries/c3-outcome.sql
```

(index)	outcome	winner	eliminator	outcome_by_runs	outcome_by_wickets	decided_by_bowl_out
0	'tie'	null	'New Zealand'	null	null	false

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

C3 passes. Four innings with correct ordinals and super-over flags, correct reversed batting order, 261 deliveries, and the tie-plus-eliminator outcome stored correctly with `fixture_winner_ck` satisfied.

On to C4, which needs a candidate first — and I want to see the source shape before ingesting, because of the key-indexing question.

Step 6 — create `scripts/find_miscounted.py` (repository root, alongside the super-over script):

```
python
```

```

import json
from pathlib import Path

DATA = Path("data/cricsheet/matches")
hits = []

for p in sorted(DATA.rglob("*.json")):
    try:
        with p.open(encoding="utf-8") as f:
            d = json.load(f)
    except Exception as e:
        print(f"UNREADABLE {p.name}: {e}")
        continue
    if not isinstance(d, dict):
        continue
    if any(i.get("miscounted_overs") for i in d.get("innings", [])):
        hits.append((p, d))

print(f"miscounted-over matches: {len(hits)}")

for p, d in hits[:5]:
    print(f"\n--- {p.name} ---")
    for idx, i in enumerate(d.get("innings", [])):
        mc = i.get("miscounted_overs")
        if not mc:
            continue
        print(f"innings ordinal {idx}: team={i.get('team')!r}")
        print(f" miscounted_overs raw: {json.dumps(mc)}")
        for key, detail in mc.items():
            balls = detail.get("balls") if isinstance(detail, dict) else detail
            print(f" key={key!r} ({type(key).__name__}) "
                  f"balls={balls!r} ({type(balls).__name__})")
            for o in i.get("overs", []):
                if str(o.get("over")) == str(key):
                    print(f" over field {o.get('over')!r} -> "
                          f"{len(o.get('deliveries', []))} deliveries")
            for o in i.get("overs", []):
                if str(o.get("over")) == str(int(key) - 1):
                    print(f" over field {int(key)-1} -> "
                          f"{len(o.get('deliveries', []))} deliveries")

```

Run it:

powershell

python scripts/find_miscounted.py

Paste the output. The count should be 175. What decides the off-by-one question is the two lines under each key: a miscounted over holds five or seven legal balls, and the delivery count in the matching over should reflect that. If the count lines up under `int(key) - 1` rather than under `key`, the map is one-based and `ingest-match.ts` is storing the wrong over number.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> python scripts/find_miscounted.py
C:\Users\BenSwartz\AppData\Local\Programs\Python\Python312\python.exe:
can't open file 'C:\\Users\\BenSwartz\\Sport-Analytics-
Tool\\scripts\\find_miscounted.py': [Errno 2] No such file or directory
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

The file wasn't created. I gave you the contents but not a command to write them, and `find_super_overs.py` you presumably made by hand.

Create it in your editor at `C:\Users\BenSwartz\Sport-Analytics-Tool\scripts\find_miscounted.py` with the contents from my previous message, save, then run:

```
powershell

python scripts/find_miscounted.py
```

If you'd rather not paste into an editor, confirm and I'll give you a PowerShell here-string that writes the file — but the script has quotes and braces throughout, and PowerShell here-strings and Python quoting interact badly enough that the editor route is less trouble.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> python scripts/find_miscounted.py
miscounted-over matches: 165
--- 1462921.json ---
innings ordinal 0: team='Uganda'
miscounted_overs raw: {"16": {"balls": "5"}}
key='16' (str) balls='5' (str)
over field 16 -> 5 deliveries
over field 15 -> 7 deliveries
--- 584928.json ---
innings ordinal 0: team='Bangladesh'
miscounted_overs raw: {"1": {"balls": 7}}
key='1' (str) balls=7 (int)
over field 1 -> 7 deliveries
over field 0 -> 7 deliveries
--- 1399114.json ---
innings ordinal 1: team='Hong Kong'
miscounted_overs raw: {"8": {"balls": 7, "umpire": "Masudur Rahman"}}
key='8' (str) balls=7 (int)
over field 8 -> 7 deliveries
```

```
over field 7 -> 6 deliveries
--- 1472524.json ---
innings ordinal 0: team='West Indies'
miscounted_overs raw: {"11": {"balls": 7, "umpire": "LS Reifer"}}
key='11' (str) balls=7 (int)
over field 11 -> 7 deliveries
over field 10 -> 6 deliveries
--- 1161233.json ---
innings ordinal 0: team='Pakistan'
miscounted_overs raw: {"13": {"balls": "7"}}
key='13' (str) balls='7' (str)
over field 13 -> 7 deliveries
over field 12 -> 6 deliveries
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Show more

Validated data pattern consistency across miscounted delivery records

The key is zero-based and matches `over.over` directly. In 1399114, key `8` has 7 deliveries while over 7 has 6; same pattern in 1472524 and 1161233. If the map were one-based, those would have to point at overs holding fewer balls than stated, which they don't. So `Number(over)` in `ingest-match.ts` is correct — my off-by-one concern was unfounded.

Three things the output settles or raises:

The string/integer inconsistency is real and confirmed: `"balls": "5"` in 1462921, `"balls": 7` in 1399114. `Number(...)` handles both. That's C4's coercion case.

The count is 165, not the 175 in your handover. Mine scanned `data/cricsheet/matches` only. Could be my narrower root, could be that 175 came from a different corpus state or a different definition. Not important to C4, but if 175 appears in `docs/database/schema.md` or ADR-003 as a justification for `innings_miscounted_over`, the figure needs correcting or reconciling — a marker will not check it, but a wrong number in a design justification is the sort of thing that undermines the rest.

`miscounted_overs` can carry an `umpire` field (1399114, 1472524) and `innings_miscounted_over` has no column for it, so it's silently dropped. Small, and arguably out of scope, but it's a source field going nowhere and you'd want that to be a decision rather than an oversight.

Step 7 — ingest a miscounted-over match with a string ball count, since that's the case that would break a naive parse:

```
powershell
```

```
npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/c3-outcome.sql
```

— ignore that, wrong command. Run:

```
powershell
```

```
ls data -Recurse -Filter 1462921.json | Select-Object -ExpandProperty FullName
```

then ingest using the path it prints:

```
powershell
```

```
npm run db:ingest --workspace=@sport-analytics/backend -- "<full path>"
```

Paste both. Expect a new `fixture_id` and the delivery count for that match. Then we verify the stored `innings_miscounted_over` row against the source, and repeat with 1399114 for the integer case.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> ls data -Recurse -Filter 1462921.json
| Select-Object -ExpandProperty FullName
C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\afrika-
continental-cup\1462921.json
```

Step 7b — ingest it:

```
powershell
```

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

Paste the output.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:ingest --
workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\afrika-continental-cup\1462921.json"
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\afrika-continental-cup\1462921.json
Ingested 1462921
  fixture_id    : 4
  people resolved : 26
  deliveries added : 216
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Ingested. Now verify the miscounted-over row landed with the right over number and a correctly coerced integer ball count, and that the stored count agrees with the deliveries actually in that over.

Step 8 — create `apps/backend/scripts/queries/c4-miscounted.sql`:

sql

```
SELECT f.source_ref,  
       i.ordinal AS innings_ordinal,  
       t.name AS batting_team,  
       m.over_number,  
       m.balls AS recorded_balls,  
       (SELECT count(*)  
        FROM delivery d  
        WHERE d.innings_id = i.innings_id  
              AND d.over_number = m.over_number  
              AND d.superseded_at IS NULL) AS delivery_rows  
FROM innings_miscounted_over m  
JOIN innings i ON i.innings_id = m.innings_id  
JOIN fixture f ON f.fixture_id = i.fixture_id  
JOIN team t ON t.team_id = i.batting_team_id  
ORDER BY f.source_ref, i.ordinal, m.over_number;
```

Run it:

powershell

```
npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/c4-miscounted.s
```

Paste the table. Expected: one row, source_ref 1462921, innings ordinal 0, Uganda, over_number 16, recorded_balls 5, delivery_rows 5. `recorded_balls` coming back as `5` rather than erroring is the string-coercion evidence C4 needs.

Note `recorded_balls` and `delivery_rows` agreeing is not guaranteed in general — the source count is legal balls while delivery rows include wides and no-balls, so a miscounted over containing an extra would differ legitimately. Here the source shows exactly 5 deliveries in over 16, so they should match. If they diverge on this match, something is wrong; on a later match, check for extras before concluding that.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --
workspace=@sport-analytics/backend -- scripts/queries/c4-miscounted.sql
> @sport-analytics/backend@0.1.0 db:sql
> tsx --env-file=.env scripts/sql.ts scripts/queries/c4-miscounted.sql
```

(index)	source_ref	innings_ordinal	batting_team	over_number	recorded_balls	delivery_rows
0	'1462921'	0	'Uganda'	16	5	'5'

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

String coercion works — `"5"` stored as smallint `5`, over number correct, delivery rows agree.

Step 9 — the integer case, 1399114. It's the better test of the two: `"balls": 7` as an integer, and it carries the `umpire` field that has nowhere to go. Also innings ordinal 1 rather than 0, so it exercises the innings lookup.

powershell

```
ls data -Recurse -Filter 1399114.json | Select-Object -ExpandProperty FullName
```

then ingest with the path it prints, then re-run the same C4 query:

powershell

```
npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/c4-miscounted.s
```

Paste all three outputs. Expected second row: 1399114, innings ordinal 1, Hong Kong, over 8, recorded_balls 7, delivery_rows 7.

If that passes, C4 is done and both remaining ingestion criteria on #27 are closed. What's then left before merge is D2 — and having three fixtures loaded rather than one makes the per-delivery storage figure meaningfully better than the estimate ADR-003 currently carries.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> ls data -Recurse -Filter 1399114.json |
Select-Object -ExpandProperty FullName
C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\asian-games-
men-s-cricket-competition\1399114.json
```

Step 9b — ingest, then verify:

powershell

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

powershell

```
npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/c4-miscounted.s
```

Paste both.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:ingest --
workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\asian-games-men-s-cricket-
competition\1399114.json"
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\asian-games-men-s-cricket-competition\1399114.json
Ingested 1399114
  fixture_id   : 5
  people resolved : 28
  deliveries added : 241
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --
workspace=@sport-analytics/backend -- scripts/queries/c4-miscounted.sql
> @sport-analytics/backend@0.1.0 db:sql
> tsx --env-file=.env scripts/sql.ts scripts/queries/c4-miscounted.sql
```

(index)		source_ref	innings_ordinal	batting_team	over_number	recorded_balls	delivery_rows
0		'1399114'	1	'Hong Kong'	8	7	'7'
1		'1462921'	0	'Uganda'	16	5	'5'

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

C4 passes. Both ball-count types coerce correctly, over numbers are right in both the zero-based innings-0 case and the innings-1 case, and delivery rows agree with recorded balls.

Where #27 stands: C3 and C4 closed, D2 outstanding, nav check outstanding, merge outstanding. Four fixtures now in the shared database — 729307, 423788, 1462921, 1399114 — totalling 955 deliveries, which supersedes the "237 deliveries, one fixture" line in your handover and is worth telling the team, since anyone else's counts will have moved under them.

Two open items from this session that need to become decisions before merge, not after:

1. **ball_number** carries no information. It's always the computed **over.position**, so it cannot exhibit the 20.4% repetition that justifies your natural key. Either populate it with the legal-ball number a scorecard prints, or drop the column.

2. `miscounted_overs.umpire` is silently discarded. Add a column or record the exclusion.

Neither is D2. Say which you want next — D2, or clearing those two first. My preference is those two, because both may touch the migration and the migration is still amendable only while the PR is unmerged, whereas D2 touches only ADR-003.

you choose, then tell me what to do

`ball_number` first. It's the one that touches the migration, and it's the one a marker can actually catch: your schema documentation justifies the natural key with a claim the database contradicts.

Before deciding what to do with the column, I want to know how bad the claim is. Your handover says the printed ball number repeats in 20.4% of overs and one over holds nineteen deliveries with six labelled `5.1`. Neither of those is reproducible from `ball_number` as currently populated, but I also can't tell from here what "printed ball number" meant when you measured it — Cricsheet doesn't publish one, so the figure must have come from a computation you did over the source, and I need to know which computation before I can tell you what the column should hold.

Step 10 — find where the claim is recorded. From the repository root:

powershell

```
Select-String -Path docs\**\*.md,evidence\**\*.md -Pattern "20.4|ball_number|nineteen|
```

Paste the output. What I'm looking for is the exact wording in `docs/database/schema.md` and ADR-003, so that whichever way we go — populate or drop — the documentation and the schema end up saying the same thing.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Select-String -Path
docs\**\*.md,evidence\**\*.md -Pattern "20.4|ball_number|nineteen|5\1" -Context
2,2
docs\database\schema.md:22:| # | Property | Consequence for the schema |
docs\database\schema.md:23:|---|---|---|
> docs\database\schema.md:24:| O1 | The printed ball number repeats within an
over in 20.4% of overs (104,818 of 514,380). One over holds nineteen deliveries,
six of them labelled 5.1. | Delivery identity is the position in the source
array, never the printed number. |
docs\database\schema.md:25:| O2 | 168 names map to more than one player
identifier, and 40 identifiers map to more than one name. Abdul Rahman is
three different people. | Players key on the registry identifier. Names are
display text and never a join key. |
docs\database\schema.md:26:| O3 | Extras types co-occur on a single delivery —
a wide with byes. Five types appear: wides, leg byes, no-balls, byes and
```

```

penalty. | Extras are separate nullable columns, not a type and a count. |
docs\database\schema.md:38:| position | ball number | extras |
docs\database\schema.md:39:|---|---|---|
> docs\database\schema.md:40:| 0 | 5.1 | no-ball |
> docs\database\schema.md:41:| 1 | 5.1 | no-ball |
> docs\database\schema.md:42:| 2 | 5.1 | no-ball |
> docs\database\schema.md:43:| 3 | 5.1 | no-ball |
> docs\database\schema.md:44:| 4 | 5.1 | no-ball |
> docs\database\schema.md:45:| 5 | 5.1 | no-ball |
> docs\database\schema.md:46:| 6 | 5.1 | — |
docs\database\schema.md:47:
docs\database\schema.md:48:Only the array position distinguishes them.
evidence\validation\729307-published-figures.md:5:
evidence\validation\729307-published-figures.md:6:**Match:** Kolkata Knight
Riders v Kings XI Punjab, 15th match, Pepsi Indian
> evidence\validation\729307-published-figures.md:7:Premier League 2014
> evidence\validation\729307-published-figures.md:8:**Date:** 26 April 2014
evidence\validation\729307-published-figures.md:9:**Venue:** Sheikh Zayed
Stadium, Abu Dhabi
evidence\validation\729307-published-figures.md:10:**Cricsheet identifier:**
729307
evidence\validation\729307-published-figures.md:14:
evidence\validation\729307-published-figures.md:15:- ESPNcricinfo full
scorecard:
> evidence\validation\729307-published-figures.md:16:
https://www.espn.cricinfo.com/series/pepsi-indian-premier-league-2014-695871/kolkata-knight-riders-vs-kings-xi-punjab-15th-match-729307/full-scorecard
evidence\validation\729307-published-figures.md:17:- Wisden live scores:
> evidence\validation\729307-published-figures.md:18:
https://www.wisden.com/team/punjab-kings-1107/live-cricket-scores/kolkata-knight-riders-vs-kings-xi-punjab-match-t20-krkp04262014175886
evidence\validation\729307-published-figures.md:19:
evidence\validation\729307-published-figures.md:20:The ESPNcricinfo URL
contains the identifier 729307, confirming that the
evidence\validation\729307-published-figures.md:57:| 1 | 13 | 2.4 | Manish
Pandey | MK Pandey |
evidence\validation\729307-published-figures.md:58:| 2 | 19 | 4.1 | Gautam
Gambhir | G Gambhir |
> evidence\validation\729307-published-figures.md:59:| 3 | 19 | 5.1 | Jacques
Kallis | JH Kallis |
evidence\validation\729307-published-figures.md:60:| 4 | 50 | 11.1 | Chris
Lynn | CA Lynn |
evidence\validation\729307-published-figures.md:61:| 5 | 59 | 12.3 | Yusuf
Pathan | YK Pathan |
PS C:\Users\BenSwartz\Sport-Analytics-Tool>

```

Show more

That resolves it, and the news is better than I thought.

The O1 table and the worked example both say the same thing: the printed ball number is `over.legal_ball_count`, computed — the six no-balls all print `5.1` because none of them advances the legal count, and the seventh delivery is also `5.1` because it's the first legal ball of over 5. And `evidence/validation/729307-published-figures.md` shows the same convention in use against a real scorecard: fall of wickets at `2.4`, `4.1`, `5.1`, `11.1`, `12.3`.

So `ball_number` is not a redundant column. It is the scorecard-facing label, it does repeat, and it is the thing that makes O1 demonstrable. The defect is only in how `ingest-match.ts` populates it: ``${over.over}.${position + 1}`` is the array position, which never repeats, so the column currently holds a value that contradicts the very property it exists to record. Fixing the population makes the schema documentation true; dropping the column would leave you unable to answer "which ball was that wicket" without recomputing on read, and your fall-of-wickets validation depends on it.

No migration change is needed. The column stays as `text NOT NULL`. This is an ingestion fix, which is cheaper — but it does mean the four loaded fixtures hold wrong values and must be reloaded.

Step 11 — fix the population in `apps/backend/scripts/ingest-match.ts`.

You need a per-over legal-ball counter. A delivery is illegal — does not advance the count — if it carries `wides` or `noballs` in extras. Byes, leg byes and penalty runs are all legal deliveries.

Find this line inside the per-over loop:

```
ts

    for (const [position, delivery] of (over.deliveries as Delivery[]).entries()) {
      sequence += 1;
      const extras = delivery.extras ?? {};
```

and replace it with:

```
ts

    let legalBalls = 0;

    for (const [position, delivery] of (over.deliveries as Delivery[]).entries()) {
      sequence += 1;
      const extras = delivery.extras ?? {};

      // The printed ball number counts legal deliveries only. Wides and
      // no-balls do not advance it, so it repeats within an over. It is a
      // label, never an identifier.
      const isLegal = extras.wides === undefined && extras.noballs === undefined;
      if (isLegal) legalBalls += 1;
      const ballNumber = `${over.over}.${Math.max(legalBalls, 1)}`;
```

Then find, in the delivery INSERT's parameter array:

ts

```
delivery.actual_delivery ?? `${over.over}.${position + 1}`,
```

and replace it with:

ts

```
ballNumber,
```

Two things I'm deliberately doing, tell me if you disagree:

`Math.max(legalBalls, 1)` handles an over opening with an illegal delivery — before any legal ball, the count is 0, and no scorecard prints `5.0`. Six consecutive no-balls followed by a legal ball then all print `5.1`, which is exactly the documented case.

I've dropped `delivery.actual_delivery`. It isn't a Cricsheet field as far as this corpus shows, and keeping a fallback for a field that never appears hides which branch is live. If you know it does appear somewhere, say so and I'll put it back.

Make the edit and tell me when it's saved. Next step reloads the four fixtures and checks that over 5 of the nineteen-delivery over reproduces the documented labels.

done

The four loaded fixtures still hold the old labels, and `ingest-match.ts` is idempotent — re-running it inserts nothing, so a reload alone won't correct them. You need to clear those rows first.

Before deciding how, I want to check what the shared database actually contains, because dropping and remigrating would destroy anyone else's work.

Step 12 — create `apps/backend/scripts/queries/state.sql`:

sql

```
SELECT f.fixture_id,
       f.source_ref,
       count(d.delivery_id) AS live_deliveries
FROM fixture f
JOIN innings i ON i.fixture_id = f.fixture_id
LEFT JOIN delivery d
      ON d.innings_id = i.innings_id
      AND d.superseded_at IS NULL
GROUP BY f.fixture_id, f.source_ref
ORDER BY f.fixture_id;
```

Run it:

```
powershell
```

```
npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/state.sql
```

Paste the output. I expect five fixtures — 729307, 423788, 1462921, 1399114, and whatever occupied the id that went missing between 1 and 3 — but if a teammate has loaded something of their own, the reset plan changes and I'd rather see it than assume the database is yours alone.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --  
workspace=@sport-analytics/backend -- scripts/queries/state.sql  
> @sport-analytics/backend@0.1.0 db:sql  
> tsx --env-file=.env scripts/sql.ts scripts/queries/state.sql
```

	(index)	fixture_id	source_ref	live_deliveries
	0	'1'	'729307'	'237'
	1	'3'	'423788'	'261'
show more	2	'4'	'1462921'	'216'
	3	'5'	'1399114'	'241'

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Four fixtures, all yours, 955 deliveries. Nobody else's work is in there. Fixture 2 was consumed by a failed attempt and rolled back — that's the sequence gap, and it's expected behaviour for an identity column.

The clean reset is a full down-and-up migration rather than deleting fixtures, because your migration is documented as rolling back and reapplying cleanly, and exercising that again is worth something in itself. It also resets the identity sequences, so the gap disappears and fixture ids become 1-4.

Step 13 — roll back and reapply:

```
powershell
```

```
npm run db:migrate:down --workspace=@sport-analytics/backend
```

```
powershell
```

```
npm run db:migrate --workspace=@sport-analytics/backend
```

Paste both outputs. If `db:migrate:down` only steps back one migration and you have more than one, tell me what it reports rather than running it repeatedly — I'd rather see the

migration list than guess at the script's semantics.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm
run db:migrate --workspace=@sport-analytics/back
end > @sport-analytics/
backend@0.1.0 db:migr...
```

PASTED

Migration reapplied cleanly. You've pasted only the `up` — I'll assume `db:migrate:down` ran first and succeeded, since `up` wouldn't have had anything to do otherwise.

Note what the migration itself says about `ball_number`: *"Display only. Never unique, and never used to join."* That is now true of the data as well as the column comment, which it wasn't before the fix.

Step 14 — reload all four fixtures. Run these in order:

powershell

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

powershell

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

powershell

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

For 729307 you'll need its path — find it with:

powershell

```
ls data -Recurse -Filter 729307.json | Select-Object -ExpandProperty FullName
```

then ingest it too.

Paste the delivery counts. They should be 261, 216, 241 and 237 again — if any differs, the `ball_number` edit changed something it shouldn't have, and we stop there.


```
deliveries added : 216
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:ingest --
workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\asian-games-men-s-cricket-competition\1399114.json"
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\asian-games-men-s-cricket-competition\1399114.json
Ingested 1399114
  fixture_id    : 3
  people resolved : 28
  deliveries added : 241
PS C:\Users\BenSwartz\Sport-Analytics-Tool> ls data -Recurse -Filter 729307.json |
Select-Object -ExpandProperty FullName
C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-
league\729307.json
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:ingest --
workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\asian-games-men-s-cricket-competition\1399114.json"
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\asian-games-men-s-cricket-competition\1399114.json
Ingested 1399114
  fixture_id    : 3
  people resolved : 28
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Show more

Three loaded, counts unchanged (261, 216, 241), so the `ball_number` edit didn't disturb delivery insertion. The last command re-ran 1399114 instead of 729307 — an incidental idempotency re-proof, which is fine, but 729307 is still not loaded.

Step 15 — load 729307:

powershell

```
npm run db:ingest --workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-An
```

Expect `fixture_id : 4` and 237 deliveries. Note the ids are not the order I suggested — 729307 lands on 4, not 1 — so anything in your validation evidence that hard-codes a fixture id needs checking. Better still, `db:validate` and your queries should select on `source_ref` rather than `fixture_id`; ids are not stable across a reset and shouldn't appear in committed evidence as if they were.

Paste the output, then we check the ball numbers reproduce the documented `5.1` case.

```
npm run db:ingest --workspace=@sport-analytics/backend --
"C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\indian-premier-league\729307.json"
```

That's the command, not its output — nothing came back. Run it and paste what the terminal prints.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:ingest --
workspace=@sport-analytics/backend -- "C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\indian-premier-league\729307.json"
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-
Tool\data\cricsheet\matches\indian-premier-league\729307.json
Ingested 729307
  fixture_id    : 5
  people resolved : 27
  deliveries added : 237
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

All four loaded, 955 deliveries. `fixture_id` 5 rather than 4 — the failed 1399114 no-op still consumed a sequence value, which reinforces the point that fixture ids are not stable and evidence should select on `source_ref`.

Now the test that matters: does `ball_number` reproduce the documented behaviour.

Step 16 — create `apps/backend/scripts/queries/ball-number.sql`:

```
sql

-- Does the printed ball number repeat within an over, as 01 claims?
SELECT f.source_ref,
       i.ordinal AS innings_ordinal,
       d.over_number,
       count(*) AS deliveries,
       count(DISTINCT d.ball_number) AS distinct_ball_numbers
FROM delivery_current d
JOIN innings i ON i.innings_id = d.innings_id
JOIN fixture f ON f.fixture_id = i.fixture_id
GROUP BY f.source_ref, i.ordinal, d.over_number
HAVING count(*) > count(DISTINCT d.ball_number)
ORDER BY count(*) - count(DISTINCT d.ball_number) DESC,
         f.source_ref, i.ordinal, d.over_number
LIMIT 10;
```

Run it:

```
powershell
```

```
npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/ball-number.sql
```

Paste the table.

What this decides: rows returned means the label repeats within overs, which is the property O1 asserts and which the old population could never produce. Zero rows means either the fix isn't working or these four matches happen to contain no extras in any over — the latter is implausible across 955 deliveries, so zero rows would mean the edit didn't take.

I'd also like to see the specific over the documentation cites, but that nineteen-delivery over is in some other match, not these four. We'll settle for demonstrating the property holds; reproducing that exact over would mean finding and loading it, which is more than #27 needs.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --  
workspace=@sport-analytics/backend -- scripts/queries/ball-number.sql  
> @sport-analytics/backend@0.1.0 db:sql  
> tsx --env-file=.env scripts/sql.ts scripts/queries/ball-number.sql
```

(index) source_ref		innings_ordinal over_number		deliveries distinct_ball_numbers	
0	'423788'	0	18	'9'	'6'
1	'1399114'	0	19	'8'	'6'
2	'1462921'	0	5	'8'	'6'
3	'1462921'	0	12	'8'	'6'
4	'423788'	3	0	'5'	'3'
5	'1399114'	0	13	'7'	'6'
6	'1399114'	1	2	'7'	'6'
7	'1399114'	1	4	'7'	'6'
8	'1399114'	1	5	'7'	'6'
9	'1399114'	1	10	'7'	'6'

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

The fix works, and the shape is exactly right: nine deliveries yielding six distinct labels, seven yielding six, and the super over at row 4 — five deliveries, three distinct labels, so two of its five balls were illegal. `ball_number` now repeats within overs, which is the property O1 asserts and the old population could never exhibit. `position_in_over` remains the identifier; the label is display only, as the column comment says.

Row 4 is worth noting: it's `423788` innings ordinal 3, over 0 — the super over. Ball numbering restarts at `0.1` there, which is correct, but it means `ball_number` alone is ambiguous across a fixture. Fine, since it's display-only and always read in innings context.

That closes the `ball_number` item. Remaining before merge:

1. `miscounted_overs.umpire` — decide, don't leave silent.

2. D2 — measured storage into ADR-003.
3. `docs/database/schema.md` in `mkdocs.yml` nav (criterion E4).
4. Commit everything and get an approval.

Step 17 — commit what's done so far, before it grows further. Four things are uncommitted: the `ingest-match.ts` fix, `scripts/sql.ts`, `scripts/queries/*.sql`, and the two Python finder scripts.

Check the working tree first:

```
powershell  
  
git status
```

Paste it. I want to see what's actually changed before telling you what to stage — in particular whether the `site/` build output Dean committed is polluting your status, and whether anything unexpected is sitting there.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git status  
On branch feat/27-delivery-event-migration  
Your branch is up to date with 'origin/feat/27-delivery-event-migration'.  
Changes not staged for commit:  
  (use "git add <file>..." to update what will be committed)  
  (use "git restore <file>..." to discard changes in working directory)  
    modified:   apps/backend/package.json  
    modified:   apps/backend/scripts/ingest-match.ts  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
    apps/backend/scripts/queries/  
Show more  
    apps/backend/scripts/sql.ts  
    evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-  
continuous-project-session.pdf  
    scripts/find_miscounted.py  
    scripts/find_super_overs.py  
    scripts/validate_source_totals.py  
no changes added to commit (use "git add" and/or "git commit -a")  
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Clean tree, no `site/` pollution on your branch. Two things to sort before staging.

The transcript PDF. Your conventions say transcripts go in `evidence/ai/transcripts/ben-swartz/`, so it belongs — but it goes in its own commit, not mixed with code. Evidence and implementation are separate concerns and a marker reading the history should see that.

`scripts/validate_source_totals.py` is untracked and not from this session. I don't know what it is. Check before you commit it:

```
powershell  
  
git log --oneline -3
```

powershell

```
Get-Content scripts/validate_source_totals.py -TotalCount 20
```

Paste both. If it's yours from earlier work on #27 it should go in; if it's a leftover experiment it shouldn't.

Assuming it's yours, the commits I'd make are three:

1. `fix(ingest): populate ball number from legal ball count` — `ingest-match.ts` only. This is a behaviour fix with a documented justification and deserves its own commit.
2. `chore(scripts): add ad-hoc SQL runner and corpus queries` — `sql.ts`, `package.json`, `scripts/queries/`, the two Python finders.
3. `docs(evidence): add AI transcript for 5-7 August session` — the PDF.

All three need `Refs #27`. Commits 1 and 2 need `Assisted-by: Claude-Web[Claude Opus 5]`; commit 3 does not, since the PDF is a record rather than generated content.

Answer on `validate_source_totals.py` and I'll give you the exact commands.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git log --oneline -3
b12b5d1 (HEAD -> feat/27-delivery-event-migration, origin/feat/27-delivery-event-migration) Merge branch 'main' of https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool into feat/27-delivery-event-migration
71a50fb docs(evidence): record the validation result for match 729307
000f729 feat(db): validate the event schema against a published scorecard
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Get-Content
scripts/validate_source_totals.py -TotalCount 20
""""Derive Innings totals from the source file, for comparison with a published
scorecard.""""
import json
MATCH = "data/cricsheet/matches/indian-premier-league/729307.json"
data = json.load(open(MATCH))
for index, innings in enumerate(data["innings"]):
    runs = 0
    wickets = 0
    legal_balls = 0
    extras = {}
    for over in innings["overs"]:
        for delivery in over["deliveries"]:
            runs += delivery["runs"]["total"]
            wickets += len(delivery.get("wickets", []))
            delivery_extras = delivery.get("extras", {})
            for kind, value in delivery_extras.items():
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

It's yours — the source-side counterpart to the SQL validation of 729307, from the work in `000f729`. It should be tracked. Note it also independently confirms the legal-ball rule you just implemented, so it belongs with the finder scripts.

Step 18 — three commits.

powershell

```
git add apps/backend/scripts/ingest-match.ts
git commit -F -
```

That will wait for input; type the message, then press Enter and Ctrl+Z then Enter to finish. If that's awkward, use `git commit` with no flags to open your editor instead — either way the message is:

```
fix(ingest): populate ball number from legal ball count
```

The printed ball number counts legal deliveries only. Wides and no-balls do not advance it, which is why it repeats within an over. Ingestion previously wrote the array position, a value that never repeats and so contradicted property 01 in docs/database/schema.md.

Verified over four fixtures: an over of nine deliveries yields six distinct ball numbers, and a super over of five yields three.

Refs #27

Assisted-by: Claude-Web[Claude Opus 5]

Then:

powershell

```
git add apps/backend/scripts/sql.ts apps/backend/scripts/queries apps/backend/package.
git commit
```

```
chore(scripts): add ad-hoc SQL runner and corpus inspection scripts
```

db:sql runs a SQL file against the project database, supporting schema validation and the storage measurement required by ADR-003. The Python scripts locate super-over and miscounted-over matches in the corpus and derive innings totals from the source for comparison against the database.

Refs #27

Assisted-by: Claude-Web[Claude Opus 5]

Then:

powershell

```
git add evidence/ai/transcripts/ben-swartz/  
git commit
```

```
docs(evidence): add AI transcript for the 5-7 August session
```

```
Refs #27
```

Run `git status` afterwards and paste it, so we confirm nothing was left behind.

One flag on the second message: it claims `db:sql` supports the ADR-003 measurement, which is true of intent but D2 isn't done yet. That's fine — the script exists for that purpose — but don't let the wording drift into implying the measurement has been taken.

```
# Please enter the commit message for your changes. Lines starting  
# with '#' will be ignored, and an empty message aborts the commit.  
#  
# On branch feat/27-delivery-event-migration  
# Your branch is ahead of 'origin/feat/27-delivery-event-migration' by 1 commit.  
# (use "git push" to publish your local commits)  
#  
# Changes to be committed:  
#   modified:   apps/backend/package.json  
#   new file:   apps/backend/scripts/queries/ball-number.sql  
#   new file:   apps/backend/scripts/queries/c3-outcome.sql  
#   new file:   apps/backend/scripts/queries/c3-super-over.sql  
#   new file:   apps/backend/scripts/queries/c4-miscounted.sql  
#   new file:   apps/backend/scripts/queries/state.sql  
#   new file:   apps/backend/scripts/sql.ts  
#   new file:   scripts/find_miscounted.py  
#   new file:   scripts/find_super_overs.py  
#   new file:   scripts/validate_source_totals.py  
#  
# Untracked files:  
@@@  
.git/COMMIT_EDITMSG [unix] (11:12 07/08/2026) 1,0-1 Top  
"/Sport-Analytics-Tool/.git/COMMIT_EDITMSG" [unix] 23L, 965B
```

You're in Vim, and the first commit went through (branch is ahead by 1). This is the second commit's message buffer.

Type `i` to enter insert mode, then paste:

```
chore(scripts): add ad-hoc SQL runner and corpus inspection scripts
```

db:sql runs a SQL file against the project database, supporting schema validation and the storage measurement required by ADR-003. The Python scripts locate super-over and miscounted-over matches in the corpus and derive innings totals from the source for comparison against the database.

Refs #27

Assisted-by: Claude-Web[Claude Opus 5]

Then press `Esc`, type `:wq` and press Enter.

If Vim isn't what you want for the remaining commit, set your editor first — `git config --global core.editor "code --wait"` for VS Code, or `notepad` if you'd rather. Do that after this commit, not mid-buffer.

done

Step 19 — third commit, the transcript:

```
powershell
```

```
git add evidence/ai/transcripts/ben-swartz/  
git commit
```

Message:

```
docs(evidence): add AI transcript for the 5-7 August session
```

Refs #27

Then:

```
powershell
```

```
git status  
git log --oneline -4
```

Paste both.

just give me the commit with the message god dammit

```
powershell
```

```
git add evidence/ai/transcripts/ben-swartz/  
git commit -m "docs(evidence): add AI transcript for the 5-7 August session" -m "Refs
```

Then:

```
powershell
```

```
git log --oneline -4
```

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git log --oneline -4  
e90fe52 (HEAD -> feat/27-delivery-event-migration) docs(evidence): add AI  
transcript for the 5-7 August session  
a86f875 chore(scripts): add ad-hoc SQL runner and corpus inspection scripts  
8e35222 fix(ingest): populate ball number from legal ball count  
b12b5d1 (origin/feat/27-delivery-event-migration) Merge branch 'main' of  
https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool into feat/27-delivery-  
event-migration  
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Three commits landed, none pushed yet. Push now so the PR reflects them:

```
powershell
```

```
git push
```

Then D2, which is the last substantive item on #27. Create

`apps/backend/scripts/queries/d2-storage.sql`:

```
sql
```

```
-- Measured storage, for the projection recorded in ADR-003.  
SELECT c.relname AS object,  
       pg_size_pretty(pg_total_relation_size(c.oid)) AS total,  
       pg_size_pretty(pg_relation_size(c.oid)) AS heap,  
       pg_size_pretty(pg_indexes_size(c.oid)) AS indexes,  
       pg_total_relation_size(c.oid) AS total_bytes  
FROM pg_class c  
JOIN pg_namespace n ON n.oid = c.relnamespace  
WHERE n.nspname = 'public'  
      AND c.relkind = 'r'  
ORDER BY pg_total_relation_size(c.oid) DESC;
```

Run it:

```
powershell
```

```
npm run db:sql --workspace=@sport-analytics/backend -- scripts/queries/d2-storage.sql
```

Paste the table.

Read the result with suspicion rather than scaling it straight to 3.2 million. Four fixtures is a small sample, page overhead dominates at this size, and the reference tables — `person`, `team`, `venue` — grow sublinearly while `delivery` grows linearly. The honest projection is per-delivery bytes from `delivery` and its dependent tables, plus a separate statement about the reference tables. I'll help you write that into ADR-003 once I see the numbers.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm run db:sql --  
workspace=@sport-analytics/backend -- scripts/queries/d2-storage.sql  
> @sport-analytics/backend@0.1.0 db:sql  
> tsx --env-file=.env scripts/sql.ts scripts/queries/d2-storage.sql
```

	(index)	object	total	heap	indexes	total_bytes
0		'delivery'	'328 kB'	'120 kB'	'176 kB'	'335872'
1		'delivery_wicket'	'80 kB'	'8192 bytes'	'64 kB'	'81920'
2		'fixture'	'80 kB'	'8192 bytes'	'64 kB'	'81920'
3		'innings'	'56 kB'	'8192 bytes'	'48 kB'	'57344'
4		'competition'	'48 kB'	'8192 bytes'	'32 kB'	'49152'
5		'venue'	'48 kB'	'8192 bytes'	'32 kB'	'49152'
6		'team'	'48 kB'	'8192 bytes'	'32 kB'	'49152'
7		'fixture_squad'	'48 kB'	'8192 bytes'	'32 kB'	'49152'
8		'person'	'48 kB'	'8192 bytes'	'32 kB'	'49152'
9		'person_alias'	'48 kB'	'8192 bytes'	'32 kB'	'49152'
10		'fixture_team'	'40 kB'	'8192 bytes'	'32 kB'	'40960'
11		'delivery_replacement'	'32 kB'	'8192 bytes'	'16 kB'	'32768'
12		'database_health'	'32 kB'	'8192 bytes'	'16 kB'	'32768'
13		'submission'	'32 kB'	'8192 bytes'	'16 kB'	'32768'
14		'fixture_official'	'32 kB'	'8192 bytes'	'16 kB'	'32768'
15		'official'	'32 kB'	'8192 bytes'	'16 kB'	'32768'
16		'innings_powerplay'	'32 kB'	'8192 bytes'	'16 kB'	'32768'
17		'dismissal_kind'	'32 kB'	'8192 bytes'	'16 kB'	'32768'
18		'fixture_player_of_match'	'24 kB'	'8192 bytes'	'16 kB'	'24576'
19		'app_user'	'24 kB'	'0 bytes'	'16 kB'	'24576'
20		'pgmigrations'	'24 kB'	'8192 bytes'	'16 kB'	'24576'
21		'innings_miscounted_over'	'24 kB'	'8192 bytes'	'16 kB'	'24576'
22		'delivery_wicket_fielder'	'24 kB'	'8192 bytes'	'16 kB'	'24576'
23		'innings_absent'	'16 kB'	'0 bytes'	'8192 bytes'	'16384'
24		'delivery_review'	'16 kB'	'0 bytes'	'8192 bytes'	'16384'

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Extrapolated storage metrics while acknowledging measurement limitations and sample size constraints

Only one number here is trustworthy: `delivery` heap at 120 kB over 955 rows is 15 full pages, about 128 bytes per row. That's a real measurement.

Everything else is floor noise. Twenty-two of these tables sit at 8192 bytes of heap — a single page, the minimum Postgres allocates — and the index figures are the same: `delivery` carries six indexes in 176 kB, which is roughly the minimum each can occupy regardless of content. You cannot extrapolate any of it.

That matters because **ADR-003 asks for a measured benchmark, and this sample doesn't provide one.** Per-delivery heap projects to roughly 410 MB at 3.19 million deliveries, but indexes are the larger share of `delivery`'s footprint and this sample says nothing about how they grow. Writing 410 MB into ADR-003 as the projection would understate the total, and writing a computed index estimate alongside a measured heap figure blurs the line between the two — exactly the thing ADR-003 flagged as unsatisfactory the first time.

The fix is a bigger sample. A few hundred matches puts every table off its minimum page and makes both heap and index growth measurable, at maybe 25–30 MB — well inside the 500 MB limit.

Step 20 — find out whether that's practical. Time a small batch first, because `ingest-match.ts` does a round trip per delivery over the session pooler and I don't know what that costs you:

```
powershell

Measure-Command {
  Get-ChildItem data\cricsheet\matches\indian-premier-league\*.json |
    Select-Object -First 10 |
    ForEach-Object {
      npm run db:ingest --workspace=@sport-analytics/backend -- $_.FullName
    }
} | Select-Object TotalSeconds
```

Paste the elapsed time. Ten matches tells us what 300 costs. If it's under a few minutes we load the sample and get a real benchmark; if it's an hour, we record what's measurable, state plainly that the index projection is unmeasured, and put the full benchmark on #27's follow-up rather than blocking the merge on it.

does this go into terminal?

Yes — same PowerShell terminal, same directory. Paste the whole block including the braces and press Enter; PowerShell handles the multi-line block fine.

It will print ten ingest summaries and then a `TotalSeconds` value. That last number is what I need.

nothing is happening, is it stuck?

Probably not stuck, but there's a real chance it's waiting on you rather than working.

`Measure-Command` suppresses output — you won't see the ten ingest summaries, so a working run and a hung run look identical. And if the paste didn't include the closing `}`, PowerShell is sitting at a continuation prompt waiting for it.

Check the prompt. If you see `>>` at the start of the line, it wants the rest of the block: press Enter on an empty line, or type `}` then Enter.

If you see the normal `PS C:\...>` prompt back, it finished.

If there's no prompt at all, it's running. Give it two or three minutes — ten matches at roughly 240 deliveries each is 2,400 round trips over a pooled connection, and that could plausibly take that long.

Press Ctrl+C if you want to stop it. Each match commits in its own transaction, so anything already ingested stays and anything mid-flight rolls back; re-running is safe either way.

Tell me what the prompt looks like.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Measure-Command {  
>> Get-ChildItem data\cricsheet\matches\indian-premier-league\*.json |  
>> Select-Object -First 10 |  
>> ForEach-Object {  
>>   npm run db:ingest --workspace=@sport-analytics/backend -- $_.FullName  
>> }  
>> } | Select-Object TotalSeconds
```

That's the continuation prompt still waiting — PowerShell has the whole block but won't run it until you close the multi-line input. **Press Enter once on the empty line.**

It should then run silently and print `TotalSeconds`.

its stuck, i cant move the cursor

Not stuck — `Measure-Command` swallows all output, so a working run shows nothing and no prompt. You can't move the cursor because it's executing.

Give it three minutes. If `TotalSeconds` appears, we have our number.

If you'd rather see progress than wait blind, press **Ctrl+C** and run this instead — same measurement, but it prints each match as it goes:

```
powershell

$start = Get-Date
Get-ChildItem data\cricsheet\matches\indian-premier-league\*.json |
  Select-Object -First 10 |
  ForEach-Object {
    npm run db:ingest --workspace=@sport-analytics/backend -- $_.FullName
  }
"elapsed seconds: $(((Get-Date) - $start).TotalSeconds)"
```

Ctrl+C is safe here. Each match commits separately, so completed ones stay, an interrupted one rolls back, and re-running skips whatever is already loaded.

We didnt do the AI register for yesterday...

Right — and worth pausing the storage work for, because it's the one thing that has a hard consequence: a document without a declaration is not marked, and the AI policy wants repository-level attribution in the README too.

But I don't know what "the AI register" is in your repository. It isn't a name I've seen in what you've shown me — you have `evidence/ai/transcripts/ben-swartz/`, and the policy asks for README declarations and per-commit `Assisted-by` lines, but a register is a fourth thing.

Two questions before I help you fill it:

```
powershell

ls evidence/ai -Recurse | Select-Object FullName
```

Paste that. And tell me what the register is meant to record — is it a table of tool, model and purpose per document, or a running log of sessions, or something your group agreed on that isn't in the course documents?

Once I can see the existing format I can help you write yesterday's entries in it. What I'd be reconstructing from: 5 August was the Supabase ruling and ADR-003, 6 August was the schema design and migration, today is C3, C4, the ball-number fix and the storage measurement. All Claude-Web[Claude Opus 5] on the parts I was involved in — but I can't vouch for what tools you used on 5 August before this session, so those entries need to come from you rather than from me.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> ls evidence\ai -Recurse | Select-Object FullName
FullName
-----
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
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C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai...
C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Show more

Truncated again. Use `-ExpandProperty` :

powershell

```
ls evidence\ai -Recurse | Select-Object -ExpandProperty FullName
```

Eighteen entries, so there's more in there than the one transcript I knew about — quite possibly the register itself.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> ls evidence\ai -Recurse | Select-Object -ExpandProperty FullName
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\ai-usage-register.csv
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\README.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\ben-swartz
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\dean-feldman
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\gabriel-raz
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\liora-rosenberg
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\nadav-sundy
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\shayna-underslak
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\README.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\ben-swartz\2026-08-05-to-2026-08-07-continuous-project-session.pdf
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\ben-
```

```
swartz\README.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\dean-
feldman\2026-08-06-continuous-project-session.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\dean-
feldman\README.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\gabriel-
raz\README.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\liora-
rosenberg\README.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\nadav-
sundy\README.md
C:\Users\BenSwartz\Sport-Analytics-Tool\evidence\ai\transcripts\shayna-
unterslak\README.md
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Show more

There it is: `evidence/ai/ai-usage-register.csv`.

Show me its format and current contents before I write anything into it:

powershell

```
Get-Content evidence/ai/ai-usage-register.csv
```

powershell

```
Get-Content evidence/ai/README.md
```

Paste both. The README should say what each column means and who is expected to fill it, and I'd rather match the existing convention than invent one.

Worth noting while you fetch it: only you and Dean have transcripts. Four members have an empty folder with a README. If the register is similarly thin, that's a group-level Sprint 1 exposure — the AI policy applies to everyone's work, not just the people who happen to be diligent — and it belongs on Tuesday's agenda rather than in your head.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Get-Content evidence/ai/ai-usage-register.csv
Date,Team member,Tool,Model,Purpose,Brief task,Output ...
```

PASTED

Format is clear enough to work from. Nine columns, one row per member per tool per work item, quoted fields, blank lines between date groups.

Before I draft your rows, three problems in that file — the first affects what I write:

Liora's two rows are structurally broken. Her fields aren't quoted, and they contain commas ("Deployment guidance, MkDocs build commands, Cloudflare Pages setup..."). Any CSV parser will read those as extra columns and the rows will not line up. Dean's entry claims he "repaired the incomplete CSV row" — that repair didn't cover these. Not yours to fix unilaterally, but it should be flagged, because a register that doesn't parse is weak evidence.

Nadav's verification line reads "only T20/IT20 data". That is the misclassification from your handover, now written into the register as a verification claim. Whatever the downloader did, the register asserts a scoping check that rests on a wrong premise.

Your last register entry is 5 August, issue #13. All of 6 August — the schema design, the migration, the 22-assertion validation — is unregistered, plus today.

For your rows I need one thing I can't supply: **what did you use on 6 August, and was it only Claude Web?** If ADR-003 or `docs/database/schema.md` involved anything else, say so. Also tell me whether the 22-assertion validation and the migration were the same session as today's PDF — the filename spans 05 to 07, which suggests one continuous session, in which case both days point at the same transcript.

Answer those and I'll draft the rows in the file's existing style for you to check.

i only used claude web

Two rows then — one for 6 August, one for today. Both cite the same transcript, since the PDF spans 05 to 07.

Append these to `evidence/ai/ai-usage-register.csv`, after Liora's last row, with a blank line before them to match the file's date grouping:

2026-08-06,Ben Swartz,Claude Web,Claude Opus 5,Planning; technical guidance; code generation; documentation,"Issue #27: design the delivery event schema, write the migration, and validate it against a published scorecard","Event model design; docs/database/schema.md; the delivery event migration; single-match ingestion script; SQL validation assertions","Design approved by all six members on 6 August before implementation; migration applied, rolled back and reapplied cleanly; 22 assertions checked by SQL over delivery_current reading nothing from the source file; idempotent resubmission confirmed by re-running the ingestion","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request"

2026-08-07,Ben Swartz,Claude Web,Claude Opus 5,Technical guidance; code generation; review,"Issue #27: confirm super-over and miscounted-over ingestion, correct the printed ball number, and measure storage","Corpus inspection scripts; ad-hoc SQL runner and validation queries; ball-number ingestion fix","Verified each result by SQL rather than by the ingestion summary; identified that the ball number was populated with the array position, contradicting property 01 in docs/database/schema.md, and corrected it; measured storage found insufficient at four fixtures and recorded as such rather than extrapolated","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request"

Two things I've deliberately not claimed: the storage measurement is described as insufficient rather than complete, because it is; and I've left the ADR-003 update out entirely, since you haven't made it yet. If you'd rather the second row wait until D2 and the merge are done, that's defensible — but 6 August's row shouldn't wait, it's already a day late.

Commit separately from code:

```
powershell

git add evidence/ai/ai-usage-register.csv
git commit -m "docs(evidence): register AI usage for 6 and 7 August" -m "Refs #27"
```

The register is a record of your usage, so it needs no **Assisted-by** line of its own — though I did draft the wording, which is the sort of recursion the policy doesn't address. If you want to be strict, add **Assisted-by: Claude-Web[Claude Opus 5]** to that commit too. I'd add it.

nvm well come back to that

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> $start = Get-Date
>> Get-ChildItem data\cricsheet\matches\indian-premier-league\*.json |
>> Select-Object -First 10 |
>> ForEach-Object {
>>   npm run db:ingest --workspace=@sport-analytics/backend -- $_.FullName
>> }
>> "elapsed seconds: $(((Get-Date) - $start).TotalSeconds)"
> @sport-analytics/backend@0.1.0 db:ingest
```

```
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082591.json
Ingested 1082591
  fixture_id    : 6
  people resolved : 28
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082592.json
Ingested 1082592
  fixture_id    : 7
  people resolved : 27
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082593.json
Ingested 1082593
  fixture_id    : 8
  people resolved : 27
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082594.json
Ingested 1082594
  fixture_id    : 9
  people resolved : 27
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082595.json
Ingested 1082595
  fixture_id    : 10
  people resolved : 27
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082596.json
Ingested 1082596
  fixture_id    : 11
  people resolved : 28
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082597.json
Ingested 1082597
  fixture_id    : 12
  people resolved : 28
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
```

```
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082598.json
Ingested 1082598
  fixture_id    : 13
  people resolved : 27
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082599.json
Ingested 1082599
  fixture_id    : 14
  people resolved : 27
  deliveries added : 0
  (already present: this run was a no-op, which is the intended behaviour)
> @sport-analytics/backend@0.1.0 db:ingest
> tsx --env-file=.env scripts/ingest-match.ts C:\Users\BenSwartz\Sport-Analytics-Tool\data\cricsheet\matches\indian-premier-league\1082600.json
Ingested 1082600
  fixture_id    : 25
  people resolved : 27
  deliveries added : 239
elapsed seconds: 676.9377405

Show more
```

Stop. Something is wrong and it needs diagnosis before any more loading.

Nine of ten matches report "already present" and inserted zero deliveries — but they were never ingested. Your earlier `Measure-Command` block wasn't hung; it ran to completion silently and loaded these nine. So they were present. That part is consistent.

But `fixture_id` jumped from 14 to 25 on the tenth. Ten fixtures consumed eleven sequence values in the silent run, then this run consumed ten more no-op values (15–24) before landing on 25. That second part is expected — the `INSERT ... ON CONFLICT DO NOTHING` still burns a sequence value. The gap is explained.

The real problem is different: **1082600 inserted 239 deliveries, while the other nine inserted zero.** If the silent run completed all ten, the tenth should also be a no-op. So the silent run got through nine and was cut off during the tenth — meaning its transaction rolled back, correctly. That's the idempotency and atomicity property working, and it's worth noting as evidence rather than alarm.

So nothing is broken. What we now know is the number I wanted:

676 seconds for ten matches, of which nine were no-ops. A no-op still does a full pass — every delivery attempts an insert. So call it ~68 seconds per match regardless. Three hundred matches is roughly **five and a half hours**.

That settles D2. Loading a few hundred fixtures is not viable at this ingestion rate before the merge, and it shouldn't block it.

What I'd do instead — tell me if you disagree:

1. Record in ADR-003 what was actually measured: 128 bytes per delivery heap, from 955 rows. State plainly that index growth is unmeasured because every table sits at its minimum page allocation at this size, and that the 500 MB constraint therefore remains unresolved rather than resolved-by-estimate.
2. Raise the ingestion rate as its own issue. ~68 seconds per match is a per-delivery round trip; the Sprint 2 batch pipeline cannot work this way, and that's a real finding from today.
3. Merge #27 without the full benchmark, with the ADR honest about it.

Say go and I'll draft the ADR-003 wording.

go

First show me what ADR-003 currently says about storage, so the replacement matches its numbering and doesn't contradict something further up:

```
powershell
```

```
Select-String -Path evidence\decisions\ADR-003*.md -Pattern "storage|MB|GB|500|estimate|benchmark"
```

Paste the output. I'd rather amend the existing section than bolt on a new one that sits awkwardly beside the estimate it's meant to replace.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Select-String -Path evidence\decisions\ADR-003*.md -Pattern "storage|MB|GB|500|estimate|benchmark"
```

PASTED

Consequence 4 is the one to amend, and reading it changes what I'd write.

It says the T20 international corpus alone is 1,266,835 deliveries at approximately 500 MB with indexes — 414 bytes per delivery, all-in. Your measured heap is 128 bytes. If that estimate is roughly right, indexes and dependent tables are more than double the heap, which is plausible for six indexes on **delivery**, but it means the measurement you have covers under a third of the footprint and cannot confirm or refute the estimate.

It also says the decision **must be recorded before ingestion begins**. Ingestion has begun — fourteen fixtures are in the shared database. Small scale, but the ADR's own condition has been crossed, and that's worth stating rather than quietly leaving.

Proposed replacement for consequence 4. Amend, don't append:

markdown

4. The agreed competition scope covers eleven franchise competitions plus men's and women's T20 internationals. The T20 international corpus alone is 1,266,835 deliveries, which is approximately 500 MB once normalised with indexes. The full scope is likely to exceed the free plan's limit. The team must decide between paying for additional storage, reducing scope, or moving provider.

A partial measurement was taken on 7 August 2026 over four fixtures and 955 deliveries: `delivery` occupied 120 kB of heap, or approximately 128 bytes per delivery. This figure is not sufficient to confirm the estimate above. Every other table sat at the minimum single-page allocation Postgres makes, and index sizes were at their floor rather than proportional to their contents, so neither index growth nor the dependent tables can be projected from this sample. The measured benchmark this record requires is therefore still outstanding.

A larger sample was not taken because single-match ingestion runs at approximately 68 seconds per fixture, making a few hundred fixtures a multi-hour operation. The measurement should be repeated once batch ingestion exists.

Fourteen fixtures have been loaded for schema validation, so ingestion has begun in a limited form before this decision was recorded. The storage decision remains outstanding and no bulk ingestion may proceed until it is taken.

Three points where I've made a judgement you might overrule:

I've kept the 414-bytes-per-delivery estimate rather than replacing it with anything derived from today. It's still the only whole-footprint figure you have, and substituting a partly-measured number for it would look more rigorous than it is.

I've named the ingestion rate as the reason for not measuring further. That's honest, but it also puts a performance problem into an ADR about database hosting. Alternative is to leave it out here and carry it entirely in the new issue.

I've written the "ingestion has begun" paragraph plainly. It's mildly self-incriminating and you could argue fourteen validation fixtures aren't what the sentence meant. I think stating it is the right call — a marker comparing the ADR to the database will notice either way.

Say if you want changes; otherwise make the edit and I'll give you the commit.

done

powershell

```
git add evidence/decisions/ADR-003-database-host-connection-and-migrations.md
git commit -m "docs(adr): record the measured storage figure and its limits" -m "The m

Refs #27
Assisted-by: Claude-Web[Claude Opus 5]"
```

Then push everything:

```
powershell
git push
```

Paste the push output.

Remaining on #27 after this: the `mkdocs.yml` nav check (E4), the `miscounted_overs.umpire` decision, and the approval. The ingestion-rate issue is new work and shouldn't hold up the merge.

done

Step 21 — the nav check, criterion E4.

```
powershell
Select-String -Path mkdocs.yml -Pattern "schema|database" -Context 2,2
```

Paste the output. If `docs/database/schema.md` isn't in the nav, E4 isn't met and the page — which is where your entire O1/O2/O3 justification lives — won't appear on the published site at all.

Note Dean's register entry mentioned the repository-wide format check being blocked by `docs/database/schema.md`, and Liora's MkDocs builds run with `--strict`. Worth knowing whether that file currently builds cleanly, since a strict build failure would take the whole documentation site down rather than just omitting a page.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Select-String -Path mkdocs.yml -
Pattern "schema|database" -Context 2,2
mkdocs.yml:22: - System Architecture and Roadmap: architecture/system-
architecture.md
mkdocs.yml:23: - API: api/overview.md
> mkdocs.yml:24: - Database: database/overview.md
> mkdocs.yml:25: - Event Model: database/schema.md
```

```
mkdocs.yml:26: - Security:
mkdocs.yml:27: - Overview: security/overview.md
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

E4 is met — `database/schema.md` is in the nav as "Event Model".

Confirm it actually builds, since a strict failure would be worse than an omission:

```
powershell
python -m mkdocs build --strict
```

Paste the output. That's Liora's documented command, so it matches how the site is built in practice.

If it passes, #27's remaining items are the `miscounted_overs.umpire` decision and the approval.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> python -m mkdocs build --strict
C:\Users\BenSwartz\AppData\Local\Programs\Python\Python312\python.exe: No
module named mkdocs
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

MkDocs isn't installed locally — it's Liora's tooling, not yours. Not a defect in the repository, just a gap in your machine.

Two options. If you want to verify it yourself:

```
powershell
pip install mkdocs
```

then re-run the build. Check whether the project pins a theme or plugins first, though, because a bare `mkdocs` install will fail strict on a missing theme:

```
powershell
Get-Content mkdocs.yml
```

Paste that and I'll tell you what else to install.

If you'd rather not install a toolchain you don't otherwise need, the alternative is to ask Liora to confirm the build passes with `schema.md` in the nav. That's arguably the better route — she owns documentation deployment, and it costs you nothing. The risk is it becomes a dependency on someone else's availability with the PR open.

My preference is to install it. It's one command, it makes you able to check your own page before you commit changes to it, and the E4 criterion is on your issue rather than hers.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Get-Content mkdocs.yml
site_name: Sport Analytics Tool
site_description: Public documentation for the COMS3011A Sport Analytics Tool
repo_name: Git Push Pray / Sport Analytics Tool
repo_url: https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool
theme:
  name: material
  features:
    - navigation.tabs
    - navigation.sections
    - content.code.copy
  nav:
    - Home: index.md
    - Getting Started:
      - Local Setup: development/setup.md
      - Repository Structure: architecture/repository-structure.md
      - Testing: development/testing.md
      - Dependencies: development/dependencies.md
    - Design:
      - Architecture: architecture/overview.md
      - System Architecture and Roadmap: architecture/system-architecture.md
      - API: api/overview.md
      - Database: database/overview.md
      - Event Model: database/schema.md
      - Security:
        - Overview: security/overview.md
        - Authentication Foundation: security/authentication.md
        - Provider Comparison: security/auth-provider-comparison.md
    - Deployment:
      - Overview: deployment/overview.md
      - Azure Investigation: deployment/azure.md
    - Methodology:
      - Git Methodology: git-methodology.md
      - Project Methodology: project_methodology.md
    - AI Use: ai/usage.md
  markdown_extensions:
    - admonition
    - attr_list
    - tables
    - toc:
        permalink: true
    - pymdownx.details
    - pymdownx.highlight
    - pymdownx.superfences:
        custom_fences:
          - name: mermaid
            class: mermaid
            format: !!python/name:pymdownx.superfences.fence_code_format
  plugins:
```

```
- search
PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more
```

Material theme and PyMdown extensions. Both come with `mkdocs-material`, so one install covers it:

```
powershell

pip install mkdocs-material
```

then:

```
powershell

python -m mkdocs build --strict
```

Paste both outputs.

Two things worth noting from that config, neither blocking:

`ai/usage.md` is in the nav — that's the page the AI policy requires for repository-level tool declarations. Worth checking it names Claude Web, since your commits carry `Assisted-by` lines and the README/usage page is where the policy wants the tool declared once for the repository.

No `mkdocs.yml` pin for the docs toolchain anywhere I've seen. Liora builds with whatever she has installed; you're about to install whatever pip gives you today. Different Material versions render differently and can differ on what strict rejects. Not urgent, but a `docs/requirements.txt` would make the build reproducible — that's hers, and it belongs on Tuesday's list rather than in your PR.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> python -m mkdocs build --strict
| ⚠ Warning from the Material for MkDocs team
|
| MkDocs 2.0, the underlying framework of Material for MkDocs,
| will introduce backward-incompatible changes, including:
|
| × All plugins will stop working – the plugin system has been removed
| × All theme overrides will break – the theming system has been rewritten
| × No migration path exists – existing projects cannot be upgraded
| × Closed contribution model – community members can't report bugs
| × Currently unlicensed – unsuitable for production use
|
| Our full analysis:
|
| https://squidfunk.github.io/mkdocs-material/blog/2026/02/18/mkdocs-2.0/
INFO - Cleaning site directory
```

```
INFO - Building documentation to directory: C:\Users\BenSwartz\Sport-Analytics-Tool\site
INFO - The following pages exist in the docs directory, but are not included in the "nav"
      configuration:
      - environment.md
      - adr\0003-azure-hosting.md
      - data\cricsheet.md
      - deployment\azure-backend.md
      - deployment\azure-fronted.md
      - deployment\cloudflare_pages.md
INFO - Documentation built in 1.39 seconds
PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more
```

Build passes strict. E4 is met and `schema.md` publishes.

The warning about MkDocs 2.0 is upstream noise from the Material team, not about your project.

The informational list is worth acting on, though not by you today. Six pages exist in `docs/` and appear on no navigation path — including `data/cricsheet.md`, which is Nadav's data-source documentation from issue #28, and three deployment pages. They build but are unreachable, so as far as a marker browsing the site is concerned they don't exist. Two further things in that list:

- `adr/0003-azure-hosting.md` is the stray `docs/adr/` your conventions say to ignore in favour of `evidence/decisions/`. Two ADR numbering schemes coexisting, both containing an ADR-003 about different subjects. That will confuse someone.
- `deployment/azure-fronted.md` is misspelled.

Raise as one issue for whoever owns documentation. Not yours, not #27.

Also: `site/` just got rebuilt in your working tree. Your handover says it's tracked in the repository, so `git status` will now show 29,000 lines of churn. Check before you commit anything else:

powershell

```
git status --short | Select-Object -First 10
git status --short | Measure-Object | Select-Object Count
```

Paste both. If `site/` is dirty we need to keep it out of your commits — that mess is Dean's issue #5, not something to absorb into #27.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git status --short | Select-Object -First 10
>> git status --short | Measure-Object | Select-Object Count
```

```
M site/404.html
M site/ai/usage/index.html
M site/api/overview/index.html
M site/architecture/overview/index.html
M site/architecture/repository-structure/index.html
M site/database/overview/index.html
M site/database/schema/index.html
M site/deployment/azure/index.html
M site/deployment/overview/index.html
M site/development/dependencies/index.html
Count
-----
29
PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more
```

As expected — 29 modified files, all `site/`, all from your build. Nothing of yours is dirty.

Discard it so it can't leak into a commit:

```
powershell

git checkout -- site/
git status --short
```

The second command should print nothing. Paste whatever it gives you.

This is a decent piece of evidence for issue #5, incidentally: committed build output means anyone who builds the docs gets spurious changes, and eventually someone commits them. Worth saying so in the issue rather than just "should be gitignored" — a concrete instance is more persuasive than a principle.

Then #27 has one open item left: `miscounted_overs.umpire`. My recommendation is not to add a column. The field appears on a minority of miscounted overs, it identifies which umpire miscounted, and nothing in the platform derives anything from it. Record the exclusion in the table comment instead — but that means a migration edit, which is still possible while the PR is unmerged.

Say whether you want the column, the comment, or neither, and I'll give you the change.

you decide

Comment, not a column.

The reasoning: `innings_miscounted_over` exists to explain an irregularity, not to feed derivation — its own comment says so. Which umpire miscounted is attribution for an officiating error, and nothing in the platform consumes it. Adding a nullable `official_id`

would also mean resolving an official who has no stable identifier, on a row where the name may not match any official in `fixture_official`. That's real work for a field nobody reads. But dropping it silently is the thing I object to, because a marker or a teammate reading the ingestion cannot tell whether it was considered or missed.

Step 22 — edit the migration. In `database/migrations/20260806150357535_delivery-event-schema.sql`, find:

sql

```
COMMENT ON TABLE innings_miscounted_over IS
  'Records overs the umpire miscounted, which legitimately hold five or seven
  legal balls. The source supplies the ball count as a string in some matches
  and an integer in others; ingestion coerces it. This table explains an
  irregularity and is not a derivation input: legal balls are counted from the
  delivery rows themselves.';
```

and replace with:

sql

```
COMMENT ON TABLE innings_miscounted_over IS
  'Records overs the umpire miscounted, which legitimately hold five or seven
  legal balls. The source supplies the ball count as a string in some matches
  and an integer in others; ingestion coerces it. This table explains an
  irregularity and is not a derivation input: legal balls are counted from the
  delivery rows themselves. Some source records also name the umpire
  responsible. That name is deliberately not stored: it is attribution for an
  officiating error that nothing in the platform derives from, and officials
  carry no stable identifier by which it could be resolved reliably.';
```

Then apply it:

powershell

```
npm run db:migrate:down --workspace=@sport-analytics/backend
npm run db:migrate --workspace=@sport-analytics/backend
```

That drops the fourteen loaded fixtures. Fine — they were validation scaffolding, and you'd reload from source anyway.

Then commit:

powershell

```
git add database/migrations/20260806150357535_delivery-event-schema.sql
git commit -m "docs(db): record the deliberate exclusion of the miscounting umpire" -r
Assisted-by: Claude-Web[Claude Opus 5]"
```

Paste the migration output. After that, #27 is ready for approval.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> npm
run db:migrate:down --wo
rkspace=@sport-analytic
s/backend >> npm run db:
migrate --workspace=@...
```

PASTED

The edit inserted the new comment but left the old string literal behind it, so the file now has two literals where one belongs. My instruction said "replace" but you evidently pasted the new block above rather than over the old one.

The `down` succeeded and the `up` rolled back cleanly, so the database is empty and consistent — no partial state to untangle.

Fix: delete the orphaned literal. In `database/migrations/20260806150357535_delivery-event-schema.sql`, find this, which currently sits immediately after the new comment ends:

sql

```
'Records overs the umpire miscounted, which legitimately hold five or seven
legal balls. The source supplies the ball count as a string in some matches
and an integer in others; ingestion coerces it. This table explains an
irregularity and is not a derivation input: legal balls are counted from the
delivery rows themselves.';
```

Delete those five lines entirely. The block must end with `...resolved reliably.';` and then go straight to the `-- Deliveries` divider.

Then reapply:

powershell

```
npm run db:migrate --workspace=@sport-analytics/backend
```

Paste the tail of the output — the last few lines are enough, I don't need the full SQL echo again.

```
syntax error at or near "'Records overs the umpire miscounted, which legitimately
hold five or seven
legal balls. The source supplies the ball count as a string in some matches
and an integer in others; ingestion coerces it. This table explains an
irregularity and is not a derivation input: legal balls are counted from the
delivery rows themselves.'"
> Rolling back attempted migration ...
error: syntax error at or near "'Records overs the umpire miscounted, which
```

legitimately hold five or seven

legal balls. The source supplies the ball count as a string in some matches and an integer in others; ingestion coerces it. This table explains an irregularity and is not a derivation input: legal balls are counted from the delivery rows themselves."

at C:\Users\BenSwartz\Sport-Analytics-Tool\node_modules\pg\lib\client.js:652:17

at process.processTicksAndRejections

(node:internal/process/task_queues:104:5)

at async Object.query (file:///C:/Users/BenSwartz/Sport-Analytics-

Tool/node_modules/node-pg-migrate/dist/bundle/index.js:3392:11)

at async runner (file:///C:/Users/BenSwartz/Sport-Analytics-

Tool/node_modules/node-pg-migrate/dist/bundle/index.js:3590:5) {

length: 438,

severity: 'ERROR',

code: '42601',

detail: undefined,

hint: undefined,

position: '11042',

internalPosition: undefined,

internalQuery: undefined,

where: undefined,

schema: undefined,

table: undefined,

column: undefined,

dataType: undefined,

constraint: undefined,

file: 'scan.l',

line: '1244',

routine: 'scanner_yyerror'

}

npm error Lifecycle script `db:migrate` failed with error:

npm error code 1

npm error path C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend

npm error workspace @sport-analytics/backend@0.1.0

npm error location C:\Users\BenSwartz\Sport-Analytics-Tool\apps\backend

npm error command failed

npm error command C:\windows\system32\cmd.exe /d /s /c cross-env

NODE_EXTRA_CA_CERTS=certs/supabase-ca.crt node-pg-migrate up --envPath

.env --migrations-dir ../../database/migrations --ignore-pattern "README.md" --

reject-unauthorized true

PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more

Same error, same byte position — the file is unchanged. Either the deletion didn't save, or it removed a different occurrence.

Look at what's actually there rather than at the editor:

powershell

Select-String -Path database\migrations\20260806150357535_delivery-event-schema.sql -P

That prints the line number of every occurrence. Paste it. There should be one; if there are two, the second is the orphan and I'll give you its exact line range to delete.

nvm it wasn't saved - now it says migrations complete

Good. Schema is up with the corrected comment, database is empty.

Commit it:

```
powershell  
  
git add database/migrations/20260806150357535_delivery-event-schema.sql  
git commit -m "docs(db): record the deliberate exclusion of the miscounting umpire" -m  
Assisted-by: Claude-Web[Claude Opus 5]"  
git push
```

Then **#27** is complete on your side. What remains is not yours to do alone:

- **Get an approval and merge.** Your methodology forbids merging your own PR. The PR description should be updated to reflect what changed today — C3, C4, the ball-number fix, the storage finding — since it currently describes only the migration.
- **Reload validation fixtures if anyone needs them.** The database is empty. Your `db:validate` evidence for 729307 will fail until it's reloaded.

And four things came out of today that belong in issues, none of them yours to fix:

1. Ingestion at ~68 seconds per fixture. Blocks the Sprint 2 batch pipeline.
2. Super-over innings must be excluded from batting and bowling aggregates. This is the one with teeth — it will silently corrupt career figures if derivation is written without it.
3. Six unnavigable docs pages, two ADR-003s under different schemes, one misspelled filename.
4. Liora's two register rows don't parse as CSV.

Say if you want any of those drafted as issue text.

have we done all the documentaion merge i merge? also i can tick everryhting on the issue hey?

Three separate things there, and the answer differs for each.

Documentation: no, not all of it. `docs/database/schema.md` still describes the schema as designed on 6 August. Today changed the ingestion behaviour behind property O1 and added the umpire exclusion, and neither is reflected there. The migration comment says it; the

published page doesn't. Whether that blocks the merge is your call, but a marker reading the site sees the page, not the migration.

You cannot merge it yourself. Your git methodology, section 10: *"A developer may not merge their own Pull Request without approval from at least one other team member."* That's your own written rule and the Sprint 1 rubric marks Git methodology on evidence of use. Merging your own PR two weeks before submission would be visible in the history and would undercut a document you wrote. Get Shayna or Dean to review.

Ticking everything on the issue: C3, C4 and E4 yes — those are demonstrated. **D2 no.** D2 asks for a measured storage benchmark and you have a partial heap figure over four fixtures, with the ADR now saying in writing that the benchmark remains outstanding. Ticking it would contradict the record you just committed.

What I'd do with D2: leave it unticked, note in the issue that the measurement was attempted and why it was insufficient, and carry it into the ingestion-performance issue. An honestly unticked criterion with a documented reason reads better than a tick that the ADR contradicts.

Want me to draft the PR description update and the issue comment?

yes

Issue #27 comment — post this, then tick C3, C4 and E4 but not D2:

markdown

Progress, 7 August

****C3 – super over: complete.**** Match 423788 (New Zealand v Australia) ingests as four innings. Ordinals 2 and 3 carry ``is_super_over = true``, over numbering restarts at 0, and the batting order reverses relative to the main innings – Australia bats the super over having bowled the second innings. Nothing infers the batting side from innings parity. The fixture stores ``outcome = 'tie'`` with a null winner and New Zealand as eliminator, satisfying ``fixture_winner_ck``.

****C4 – miscounted over: complete.**** Two matches were loaded to cover both source types: 1462921 supplies the ball count as the string ``"5"``, 1399114 as the integer ``7``. Both coerce correctly, both store the correct over number, and in both cases the recorded ball count agrees with the delivery rows in that over. The map is keyed on the zero-based over number, matching ``over.over``, so no adjustment is required.

****E4 – navigation: complete.**** ``database/schema.md`` appears in ``mkdocs.yml`` under Design as `*Event Model*`, and ``mkdocs build --strict`` passes.

****D2 – storage measurement: not complete.**** A measurement was taken over four fixtures and 955 deliveries. ``delivery`` occupied 120 kB of heap, approximately 128 bytes per delivery. That figure alone cannot confirm the 414 bytes per delivery estimate in ADR-003 consequence 4, because every other table sat at the minimum single-page allocation and index sizes were at their floor rather than proportional to their contents. Neither index growth nor the dependent tables can be projected from this sample.

A larger sample was not taken because single-match ingestion runs at approximately 68 seconds per fixture, making a few hundred fixtures a multi-hour operation. ADR-003 has been updated to record the partial figure, its limits, and the fact that the required benchmark remains outstanding. The measurement should be repeated once batch ingestion exists; raised separately as an ingestion performance issue.

Additional change

Ingestion previously populated ``delivery.ball_number`` with the array position, which never repeats. The printed ball number counts legal deliveries only, so it does repeat within an over – the property recorded as 01 in ``docs/database/schema.md`` and cited in the ``position_in_over`` column comment as the reason that column is the identifier. The database therefore contradicted its own stated justification. Ingestion now derives the label from a per-over legal ball count. Verified across four fixtures: an over of nine deliveries yields six distinct ball numbers, and a super over of five yields three.

The source also names the umpire responsible on some miscounted overs. That field

is deliberately not stored, and the exclusion is now recorded in the `innings\_miscounted\_over` table comment rather than left silent.

PR description — replace the Changes, Testing and Known Limitations sections with:

markdown

Changes

- Delivery event migration: 22 tables, one view, 12 indexes.
- Single-match ingestion script and SQL validation queries.
- Ad-hoc SQL runner (`db:sql`) and corpus inspection scripts.
- Corrected `ball\_number` population to the legal ball count.
- Recorded the deliberate exclusion of the miscounting umpire.
- ADR-003 updated with the partial storage measurement and its limits.

Testing

Migration applied, rolled back and reapplied cleanly. Match 729307 validated by 22 SQL assertions over `delivery\_current`, reading nothing from the source file. Super-over and miscounted-over ingestion confirmed as described in issue #27. Idempotent resubmission confirmed: re-running ingestion adds zero rows. `mkdocs build --strict` passes.

Known Limitations

- The storage benchmark ADR-003 requires is not complete. See the issue comment.
- `docs/database/schema.md` has not been updated for the `ball\_number` correction or the umpire exclusion.
- Ingestion runs at approximately 68 seconds per fixture. Raised separately.

That last section is deliberately frank. Reviewers should know what they're approving, and your methodology requires known limitations to be documented before merge.

The `schema.md` gap is the one I'd fix rather than declare, if you have the energy. Say the word.

lets fix it now do the PR is complete, then ill tick everything from A-E barring D2

Show me the two places in `docs/database/schema.md` that need changing:

powershell

```
Select-String -Path docs\database\schema.md -Pattern "ball number|ball_number|miscount
```

Paste the output. I want to see the O1 row and the worked example table again in their current form, plus wherever `innings_miscounted_over` is described, so the edits fit the page's existing structure rather than being bolted on.

V

Looks like a stray keystroke. Run the command and paste the output:

```
powershell

Select-String -Path docs\database\schema.md -Pattern "ball number|ball_number|miscount
```

PS C:\Users\BenSwartz\Sport-Analytics-Tool> Select-String -Path docs\database\schema.md -Pattern "ball number|ball_number|miscounted" -Context ...

PASTED

Two edits. The page's O1 claim and section 2.1 line are already correct in substance — what's missing is that the database now demonstrates them, and the umpire exclusion.

Edit 1 — line 81. Replace:

```
markdown

The printed ball number is stored for display only, and never used to join.
```

with:

```
markdown

The printed ball number is stored for display only, and never used to join. It is derived at ingestion from a count of legal deliveries within the over: wides and no-balls do not advance it, which is why it repeats. Storing the array position here instead would produce a value that never repeats, and the column would no longer demonstrate the property that makes `position_in_over` the identifier.
```

Edit 2 — the numbered item at line 218. After the existing text about coercion and the table explaining an irregularity, append:

```
markdown
```

Some source records also name the umpire responsible for the miscount. That name is deliberately not stored: it is attribution for an officiating error that nothing in the platform derives from, and officials carry no stable identifier by which it could be resolved reliably.

Match the existing indentation of that numbered item — the continuation lines are indented three spaces.

One thing I'm leaving alone: O6 says 175 innings carry `miscounted_overs`, and my scan found 165 matches. Those count different things — innings versus matches — so they aren't necessarily in conflict, and I can't tell from here which is right. Not worth touching without checking, and not worth checking today.

Make both edits, then:

```
powershell

git add docs/database/schema.md
git commit -m "docs(db): document ball number derivation and the umpire exclusion" -m
Assisted-by: Claude-Web[Claude Opus 5]"
git push
```

Then update the PR description, post the issue comment, tick A through E excluding D2, and request review.

give me the new PR description

markdown

Purpose

Implements the delivery event schema approved by all six members on 6 August 2026 and documented in ``docs/database/schema.md``. Every structural decision traces to an observed property of the Cricsheet source data recorded in section 1 of that document.

Changes

- Delivery event migration: 22 tables, one view, 12 indexes.
- Single-match ingestion script and SQL validation queries.
- Ad-hoc SQL runner (``db:sql``) and corpus inspection scripts.
- Corrected ``ball_number`` population to a per-over legal ball count.
- Recorded the deliberate exclusion of the miscounting umpire.
- ADR-003 updated with the partial storage measurement and its limits.
- ``docs/database/schema.md`` updated for both of the above.

Testing

Migration applied, rolled back and reapplied cleanly on more than one occasion.

Match 729307 validated by 22 assertions in SQL over ``delivery_current``, reading nothing from the source file. All pass, including the complete ten-row fall of wickets and the exclusion of run outs from bowler credit.

Super over confirmed on match 423788: four innings, ordinals 2 and 3 flagged, over numbering restarting at 0, batting order correctly reversed, and the fixture stored as a tie with a null winner and an eliminator.

Miscounted over confirmed on matches 1462921 and 1399114, covering both source types – the ball count as a string and as an integer. Both coerce correctly and the recorded count agrees with the delivery rows in that over.

Ball number verified across four fixtures: an over of nine deliveries yields six distinct printed numbers, and a super over of five yields three.

Idempotent resubmission confirmed: re-running ingestion adds zero rows. An interrupted ingestion rolled back without partial state.

``mkdocs build --strict`` passes and ``database/schema.md`` appears in the navigation.

Known Limitations

- The measured storage benchmark ADR-003 requires is not complete. A partial figure of approximately 128 bytes per delivery of heap was taken over four fixtures, but every other table sat at the minimum page allocation and index sizes were at their floor, so neither index growth nor the dependent tables can be projected from it. ADR-003 records this and the storage decision remains

outstanding.

- Ingestion runs at approximately 68 seconds per fixture, which is why a larger sample was not taken. Raised separately as an ingestion performance issue.
- Super-over innings are stored and flagged but nothing yet excludes them from batting and bowling aggregates. Raised separately; it must be handled before derivation is written, or career figures will be wrong across 99 matches.
- The database is currently empty following the final migration cycle. Validation fixtures need reloading before `db:validate` will pass.

Related Issue

Closes #27

The third limitation is the one reviewers should actually stop on. Everything else is bookkeeping; that one is a correctness trap waiting for whoever writes derivation.

This branch is out-of-date with the base branch

Expected — main has moved while you've been working. Your methodology requires updating the branch and resolving conflicts before a PR is merged, so this is a step, not a problem.

powershell

```
git fetch origin
git merge origin/main
```

Paste the output. If it merges cleanly, push and you're done. If there are conflicts, paste the conflicted file list and we'll work through them — `docs/` and `mkdocs.yml` are the likely candidates given how much documentation activity there's been from Dean and Liora.

Do not rebase. Your methodology specifies merge commits and forbids rewriting a branch others may have reviewed.

```
Merge remote-tracking branch 'origin/main' into feat/27-delivery-event-migration
# Please enter a commit message to explain why this merge is necessary,
# especially if it merges an updated upstream into a topic branch.
#
# Lines starting with '#' will be ignored, and an empty message aborts
# the commit.
~
~
~
~
~
```

```
.git/MERGE_MSG [unix] (12:00 07/08/2026)
"/Sport-Analytics-Tool/.git/MERGE_MSG" [unix] 6L, 307B
Show more
```

Press `:wq` and Enter to accept it.

```
powershell

git push
git log --oneline -3
```

so I can move the issue into done?

The board sequence your methodology defines: the issue goes to **In Review** when the PR opens, and to **Done** only after the merge. It's in **In Review** now and should stay there until Shayna or Dean approves and merges.

1. Push the merge commit, if you haven't.
2. Ask a team member to review — say which one and when, so it isn't left open over the weekend.
3. They approve and merge.
4. You close #27, delete the branch, move the board item to **Done**.

Sprint 1 is 25 August. Eighteen days, and this PR has been open a while. Getting the review request out today matters more than anything else on the list.

they did approve and merge

Then finish the workflow. Your methodology, section 10, lists what happens after a merge:

1. **Close #27** — should be automatic if the PR body carried `Closes #27`. Check it actually closed rather than assuming.
2. **Delete the branch** in Gitea.
3. **Move the board item to Done**.
4. **Record follow-up work as new issues**. This is the step people skip and the one that costs you at marking.

Locally:

```
powershell  
  
git switch main  
git pull origin main  
git branch -d feat/27-delivery-event-migration
```

Then the four follow-ups, which are now unrecorded work sitting only in this conversation:

- **Super-over exclusion from aggregates**. The one with real consequences.
- **Ingestion performance** — 68 seconds per fixture blocks the Sprint 2 batch pipeline.
- **Storage decision** — ADR-003 consequence 4, still no owner, still no date, and now explicitly outstanding in writing.
- **Documentation hygiene** — six un navigable pages, two ADR-003s, `azure-fronted.md`, the unparseable register rows.

The first three are Sprint 2 blockers and the middle one is unowned. Say which you want drafted and I'll write them in your issue template format.

Pr closed, i think I theres wuite a few stale branches?

Look before deleting anything — some may be other people's active work.

```
powershell  
  
git fetch --prune  
git branch -r --merged origin/main
```

The first cleans up local references to branches already deleted on the server. The second lists remote branches whose work is fully in `main` — those are the genuinely stale ones, safe to delete.

Then:

```
powershell  
git branch -r --no-merged origin/main
```

Those are branches with unmerged work. Do not touch them; some will be Gabriel's or Nadav's in-flight.

Paste both lists.

Worth noting your methodology says the completed branch must be deleted after merge, and a normal branch should be completed within five working days — so a pile of stale branches is itself evidence the process isn't being followed consistently, which the Sprint 1 rubric marks. Deleting them is worth doing, but the person who should delete each one is whoever merged it, not you unilaterally. Raise it at Thursday's standup rather than tidying up other people's branches without asking.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git fetch --prune  
>> git branch -r --merged origin/main  
From https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool  
- [deleted]      (none)   -> origin/chore/13-establish-database-migrations  
- [deleted]      (none)   -> origin/docs/cloudflare-pages  
- [deleted]      (none)   -> origin/docs/cloudflare-pages-deployment  
- [deleted]      (none)   -> origin/feat/14-google-oauth-oidc-foundation  
- [deleted]      (none)   -> origin/feat/27-delivery-event-migration  
- [deleted]      (none)   -> origin/feat/28-add-cricsheet-downloader  
remote: Enumerating objects: 1, done.  
remote: Counting objects: 100% (1/1), done.  
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Unpacking objects: 100% (1/1), 316 bytes | 45.00 KiB/s, done.  
49205b0..189a665 main    -> origin/main  
origin/HEAD -> origin/main  
origin/chore/13-provision-database-access  
origin/ci/10-restore-green-ci  
origin/docs/73-align-architecture-roadmap  
origin/docs/cloudflare-pages-deployment-v2  
origin/docs/final-cloudflare-documentation-updates  
origin/feat/27-delivery-event-schema  
origin/infra/15-azure-backend-deployment  
origin/main  
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git branch -r --no-merged  
origin/main  
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Six were already deleted server-side, including yours. Seven remain, and **every one is fully merged into `main`** — the `--no-merged` list is empty, so nothing unmerged is at risk. They're

all safe to delete.

Two of them are yours: `chore/13-provision-database-access` and `feat/27-delivery-event-schema`. Delete those:

```
powershell
```

```
git push origin --delete chore/13-provision-database-access  
git push origin --delete feat/27-delivery-event-schema
```

The other five belong to Shayna (`ci/10`), Dean (`docs/73`, `infra/15`) and Liora (the two Cloudflare ones). Merged and safe, but deleting another member's branch without asking is the kind of thing that annoys people for no gain. Raise it at Thursday's standup — one line, "seven merged branches are still on the server, please delete yours" — and it becomes evidence of the methodology being followed rather than a unilateral cleanup.

Worth noting the pattern for that conversation: `feat/27-delivery-event-schema` and `feat/27-delivery-event-migration` are two branches for one issue, and Liora has three Cloudflare branches with no issue number in any of them. Your methodology requires the issue number in the branch name. Not a crisis, but the Sprint 1 rubric marks Git methodology on evidence of use, and a marker skimming branch names will see it.

what's next?

#27 is closed. What's left splits into what's yours and what isn't.

Yours, and undone: the four follow-up issues. They exist only in this conversation right now. Your methodology says incomplete or follow-up work must be recorded as a new issue after a merge — that step hasn't happened, and the super-over exclusion in particular is a correctness trap that will be expensive if someone writes derivation before it's recorded.

Not yours, but Sprint 1 risk: password reset and account deletion are named key requirements in the brief and you've seen neither addressed. The Supabase Auth compliance question — ADR-003 explicitly excluded Auth from Brendan's ruling, and nobody has asked him about it. Both are Tuesday agenda items with named owners, not things to keep carrying in a handover.

The rubric point. Sprint 1 is roughly 80% process and documentation. You now have a merged migration, an ADR, validated evidence, a documentation page and an honest limitations record — that's the 20% plus a good chunk of the 80%. What the rubric asks for that I haven't seen evidence of: stakeholder interaction records, sprint planning records, and a populated board showing movement. Your methodology specifies Tuesday meetings and Thursday standups with saved records. If those records don't exist in `evidence/`, that's a bigger mark than anything left in the database work.

My recommendation for the rest of today: draft the four issues, then check whether the stakeholder and standup records actually exist.

Which do you want first?

what about my AI register update?

Still undone — you said "nvm we'll come back to that" and we didn't. Two rows outstanding, 6 August and today.

The 7 August row needs revising, though. What I drafted earlier described the storage measurement as the last thing done; since then you also corrected `ball_number` in the documentation, recorded the umpire exclusion, and merged. Use these:

```
2026-08-06,Ben Swartz,Claude Web,Claude Opus 5,Planning; technical guidance; code generation; documentation,"Issue #27: design the delivery event schema, write the migration, and validate it against a published scorecard","Event model design; docs/database/schema.md; the delivery event migration; single-match ingestion script; SQL validation assertions","Design approved by all six members on 6 August before implementation; migration applied, rolled back and reapplied cleanly; 22 assertions checked by SQL over delivery_current reading nothing from the source file; idempotent resubmission confirmed by re-running the ingestion","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request"
```

```
2026-08-07,Ben Swartz,Claude Web,Claude Opus 5,Technical guidance; code generation; documentation; review,"Issue #27: confirm super-over and miscounted-over ingestion, correct the printed ball number, measure storage, and complete the Pull Request","Corpus inspection scripts; ad-hoc SQL runner and validation queries; ball-number ingestion fix; ADR-003 storage amendment; schema documentation updates; issue and Pull Request wording","Verified every result by SQL rather than by the ingestion summary; identified that the ball number was populated with the array position, contradicting property 01 in docs/database/schema.md, and corrected it; the storage measurement was found insufficient at four fixtures and recorded as outstanding rather than extrapolated; reviewed and merged by another team member","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request"
```

Append to `evidence/ai/ai-usage-register.csv` with a blank line before them, matching the file's date grouping.

You're on `main` now, and your git methodology prohibits committing directly to it. So this needs its own issue and branch:

```
powershell

git switch -c docs/NN-register-ai-usage-6-7-august
```

Replace `NN` with the issue number once you've created it. Then:

powershell

```
git add evidence/ai/ai-usage-register.csv
git commit -m "docs(evidence): register AI usage for 6 and 7 August" -m "Refs #NN
Assisted-by: Claude-Web[Claude Opus 5]"
git push -u origin docs/NN-register-ai-usage-6-7-august
```

The **Assisted-by** line is arguably odd on a record of your own usage, but I drafted the wording, so it's accurate.

Also still worth raising: the transcript PDF spans 05 to 07 and today's session is now longer than when you exported it. If the PDF is stale, re-export before Sprint 1 — the policy allows an unedited transcript to be requested.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git add evidence/ai/ai-usage-
register.csv
>> git commit -m "docs(evidence): register AI usage for 6 and 7 August" -m "Refs
#NN
>> Assisted-by: Claude-Web[Claude Opus 5]"
>> git push -u origin docs/NN-register-ai-usage-6-7-august
On branch docs/NN-register-ai-usage-6-7-august
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    apps/backend/scripts/queries/d2-storage.sql
    site/adr/
    site/ai_policy.pdf
    site/architecture/system-architecture/
    site/data/
    site/deployment/azure-backend/
    site/deployment/azure-fronted/
    site/deployment/cloudflare_pages/
    site/environment/
    site/security/auth-provider-comparison/
    site/security/authentication/
nothing added to commit but untracked files present (use "git add" to track)
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a new pull request for 'docs/NN-register-ai-usage-6-7-august':
remote: https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/pulls/new/docs/NN-register-ai-usage-6-7-august
remote:
remote: . Processing 1 references
remote: Processed 1 references in total
To https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool.git
 * [new branch] docs/NN-register-ai-usage-6-7-august -> docs/NN-register-ai-
usage-6-7-august
branch 'docs/NN-register-ai-usage-6-7-august' set up to track 'origin/docs/NN-
register-ai-usage-6-7-august'.
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Three problems, all fixable.

The commit didn't happen. "nothing added to commit" — you haven't made the CSV edit yet, so `git add` had nothing to stage. The push still ran and created an empty branch on the server, literally named `docs/NN-...` with the placeholder in it. I should have told you to create the issue first; that's on me.

`d2-storage.sql` **is untracked.** It was created after the earlier commit and never staged. It belongs with the others.

`site/` **has grown new untracked directories** from your MkDocs build. More evidence for Dean's issue #5.

Fix, in order:

1. Delete the misnamed branch:

powershell

```
git push origin --delete docs/NN-register-ai-usage-6-7-august
git switch main
git branch -D docs/NN-register-ai-usage-6-7-august
```

2. Create the Gitea issue for the register update, with a description and acceptance criteria as your methodology requires. Tell me the number it gets.

3. Make the CSV edit — append the two rows I gave you.

Then I'll give you the branch, add and commit commands with the real issue number, including `d2-storage.sql`.

Leave `site/` alone entirely.

whats the command to pull newest verison

powershell

```
git switch main
git pull origin main
```

it won't let me update the register bacuse it says its too old

"Too old" isn't wording git or an editor normally produces, so I don't know what's blocking you. Paste the exact message and say where it appeared — terminal, editor, or the Gitea web interface.

Two likely candidates:

If you were editing the file in the Gitea web editor, it refuses when the file has changed on the server since you opened it. Fix is to reload the page and redo the edit — but you should be editing locally on a branch anyway, not through the web interface, since your methodology routes everything through a PR.

If it's your local editor saying the file changed on disk, that's the `git pull` you just ran overwriting it underneath you. Reload the file, then redo the edit.

Either way, check what you actually have first:

```
powershell  
  
git status  
git log --oneline -3 -- evidence/ai/ai-usage-register.csv
```

Paste both, and the exact error text.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git status  
>> git log --oneline -3 -- evidence/ai/ai-usage-register.csv  
On branch main  
Your branch is up to date with 'origin/main'.  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
    apps/backend/scripts/queries/d2-storage.sql  
    site/adr/  
    site/ai_policy.pdf  
    site/architecture/system-architecture/  
    site/data/  
Show more  
    site/deployment/azure-backend/  
    site/deployment/azure-fronted/  
    site/deployment/cloudflare_pages/  
    site/environment/  
    site/security/auth-provider-comparison/  
    site/security/authentication/  
nothing added to commit but untracked files present (use "git add" to track)  
77fa904 merge: resolve conflicts with main  
c67e6af docs(testing): document automated testing foundation  
afaec61 test(config): establish automated testing commands  
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Nothing here is blocking you. You're on `main`, up to date, and the register file is clean — the only unstaged things are `d2-storage.sql` and `site/` noise. Git isn't refusing anything.

So whatever said "too old" came from somewhere else, and I can't diagnose it without the actual text. Paste the message and tell me where you saw it — was this Excel opening the CSV, the Gitea web editor, or something else?

Meanwhile, the sequence hasn't changed and step 2 is still outstanding: **create the Gitea issue for the register update and tell me its number**. Everything after that depends on it.

Failed to save 'ai-usage-register.csv': The content of the file is newer. Please compare your version with the file contents or overwrite the content of the file with your changes.

That's VS Code, and it's correct to stop you. You opened the file, then `git pull` changed it on disk underneath your editor. Your buffer is based on the old version — saving would silently discard whatever came in from `main`, and someone else has clearly been editing the register.

Don't overwrite. Close the file without saving, reopen it, and redo your edit against the current contents. Your two rows are appended at the end, so redoing them is trivial.

Before you do, check what changed — if a teammate added rows for 7 August, yours need to slot in without disturbing theirs:

powershell

```
git log --oneline -3 -- evidence/ai/ai-usage-register.csv
```

Wait — I already have that from your last paste, and none of those three commits mention the register directly. The `merge: resolve conflicts with main` at the top likely brought it in. Run this instead to see the actual diff:

powershell

```
git log -1 -p -- evidence/ai/ai-usage-register.csv | Select-Object -First 60
```

Paste it. Then reopen the file, append your rows, and we'll get the issue number and commit.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git log -1 -p -- evidence/ai/ai-usage-  
register.csv | Select-Object -First 60  
commit 77fa904d314a40c58fadc388d52bb3ffb6b25821  
Merge: c67e6af e46ded3  
Author: DeanFeldman <dean.feldman05@gmail.com>  
Date: Fri Aug 7 07:53:33 2026 +0200  
    merge: resolve conflicts with main  
  
    Assisted-by: ChatGPT-Web[GPT-5.6 Thinking]  
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

A merge commit with no diff shown means git resolved it without changes attributable to that commit — so the register content came from one of the two parents. Look at what the file actually contains now rather than chasing history:

powershell

Paste that. If a teammate has added rows since you last saw the file, your two rows go after them. If the tail is still Liora's 6 August entries, nothing changed in the register and the conflict was elsewhere in the merge.

Then reopen the file in VS Code — it will load the current version — append your rows, save, and tell me the issue number so I can give you the branch and commit.

ct-Object -Last 6 "2026-08-06","Nadav Sundy","ChatGPT Web","GPT-5.6 Thinking","Code assistance; documentation; review","Issue #28: generate and re...

PASTED

The register has been substantially rewritten since you last saw it. Every field is now quoted, including dates — Liora's malformed rows are fixed, and there's a new Dean entry for issue #72 that wasn't there before. Blank lines between date groups appear to be gone too.

So my earlier rows are in the wrong style. Use these instead — every field quoted, matching the current file:

```
"2026-08-06","Ben Swartz","Claude Web","Claude Opus 5","Planning; technical guidance; code generation; documentation","Issue #27: design the delivery event schema, write the migration, and validate it against a published scorecard","Event model design; docs/database/schema.md; the delivery event migration; single-match ingestion script; SQL validation assertions","Design approved by all six members on 6 August before implementation; migration applied, rolled back and reapplied cleanly; 22 assertions checked by SQL over delivery_current reading nothing from the source file; idempotent resubmission confirmed by re-running the ingestion","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request"
"2026-08-07","Ben Swartz","Claude Web","Claude Opus 5","Technical guidance; code generation; documentation; review","Issue #27: confirm super-over and miscounted-over ingestion, correct the printed ball number, measure storage, and complete the Pull Request","Corpus inspection scripts; ad-hoc SQL runner and validation queries; ball-number ingestion fix; ADR-003 storage amendment; schema documentation updates; issue and Pull Request wording","Verified every result by SQL rather than by the ingestion summary; identified that the ball number was populated with the array position, contradicting property 01 in docs/database/schema.md, and corrected it; the storage measurement was found insufficient at four fixtures and recorded as outstanding rather than extrapolated; reviewed and merged by another team member","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request"
```

Append to the end of the file, no blank line before them.

Note Nadav's entry still says "only T20/IT20 data" — the misclassification survived the rewrite and is now stated as a verification claim in a cleaned-up register. Worth telling him directly rather than waiting for Tuesday.

Reopen the file in VS Code so it loads the current version, append, save. Then give me the issue number.

give it to me as a terminal command

PowerShell here-strings mangle nothing if the delimiter is single-quoted — `@'` prevents any variable expansion, so the text goes in verbatim. Paste this whole block:

```
powershell
```

```
@'
```

```
"2026-08-06","Ben Swartz","Claude Web","Claude Opus 5","Planning; technical guidance;  
"2026-08-07","Ben Swartz","Claude Web","Claude Opus 5","Technical guidance; code gener  
'@ | Add-Content -Path evidence\ai\ai-usage-register.csv -Encoding utf8
```

Close the file in VS Code first — if it's open with unsaved changes, you'll hit the same conflict.

Then check it landed:

```
powershell
```

```
Get-Content evidence/ai/ai-usage-register.csv | Select-Object -Last 2
```

Paste that, and give me the issue number when you have it.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Get-Content evidence/ai/ai-usage-  
register.csv | Select-Object -Last 2  
"2026-08-06","Liora Rosenberg","ChatGPT Web","GPT-5.5","Documentation;  
deployment; Git troubleshooting; Pull Request drafting","Deploy the MkDocs  
documentation to Cloudflare Pages using Wrangler; troubleshoot Cloudflare  
authentication and Git issues; update the deployment documentation; prepare the  
Pull Request description","Deployment commands; deployment documentation; Git  
troubleshooting steps; Pull Request wording","Executed and verified all commands  
manually; successfully completed the MkDocs build using python -m mkdocs build --  
strict and the Wrangler deployment using npx wrangler pages deploy site --project-  
name=sports-analytics-tool; reviewed and adapted the documentation before  
committing it","Successful MkDocs build output; successful Wrangler deployment  
output; Cloudflare Pages URL (https://sports-analytics-tool.pages.dev); updated  
deployment documentation; Pull Request""2026-08-06","Ben Swartz","Claude
```

Web","Claude Opus 5","Planning; technical guidance; code generation; documentation","Issue #27: design the delivery event schema, write the migration, and validate it against a published scorecard","Event model design; docs/database/schema.md; the delivery event migration; single-match ingestion script; SQL validation assertions","Design approved by all six members on 6 August before implementation; migration applied, rolled back and reapplied cleanly; 22 assertions checked by SQL over delivery_current reading nothing from the source file; idempotent resubmission confirmed by re-running the ingestion","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request" "2026-08-07","Ben Swartz","Claude Web","Claude Opus 5","Technical guidance; code generation; documentation; review","Issue #27: confirm super-over and miscounted-over ingestion, correct the printed ball number, measure storage, and complete the Pull Request","Corpus inspection scripts; ad-hoc SQL runner and validation queries; ball-number ingestion fix; ADR-003 storage amendment; schema documentation updates; issue and Pull Request wording","Verified every result by SQL rather than by the ingestion summary; identified that the ball number was populated with the array position, contradicting property O1 in docs/database/schema.md, and corrected it; the storage measurement was found insufficient at four fixtures and recorded as outstanding rather than extrapolated; reviewed and merged by another team member","evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and linked Pull Request" PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more

Broken — the file had no trailing newline, so Liora's row and your 6 August row are now fused into one line. Look at `...Pull Request""2026-08-06","Ben Swartz"...` — two records, one line, and the CSV won't parse.

Undo and redo:

```
powershell  
git checkout -- evidence/ai/ai-usage-register.csv
```

Then append a newline first, then the rows:

```
powershell  
Add-Content -Path evidence\ai\ai-usage-register.csv -Value "" -NoNewline
```

That won't do it either — cleaner to write the newline explicitly:

```
powershell  
[System.IO.File]::AppendAllText("$PWD\evidence\ai\ai-usage-register.csv", "`r`n")
```

Then re-run the same `@'...'@ | Add-Content` block from my previous message.

Then verify:

powershell

```
(Get-Content evidence/ai/ai-usage-register.csv | Measure-Object).Count  
Get-Content evidence/ai/ai-usage-register.csv | Select-Object -Last 3
```

The last three lines should be Liora's row, then each of yours, on separate lines. Paste it.

im not on a new branch

Correct — you're on `main`, and your methodology prohibits committing there. But nothing is committed yet, so there's no problem: the working-tree change moves with you when you switch branches.

Order matters, though. Fix the file first, then branch, then commit — switching branches with a half-broken file just carries the breakage along.

So:

1. Run the `git checkout --` and the two append commands from my last message.
2. Verify the last three lines are separate.
3. **Create the Gitea issue** and tell me the number.

Then I'll give you the branch and commit commands with the real number in them.

Everything is blocked on that issue number, which is the one step you keep skipping past.

We decided we are going to make a folder each for the registers because we keep hitting conflicts

Sensible — a single append-only CSV touched by six people conflicts constantly, and you've hit it twice today.

But it changes what you should do right now, and I don't know the agreed shape. Before writing anything:

What's the format? One CSV per member (`evidence/ai/registers/ben-swartz.csv`) with the same nine columns, or something else? And is there a combining step — a script that concatenates them for submission — or does the group register cease to exist?

Who's doing the migration? Splitting the existing file means moving fourteen-odd rows into six files, and that's one person's job, not six people each doing their bit. If it's already assigned, your two rows go into whatever the new structure is. If it isn't, appending to the old file today is still correct and you migrate with everyone else.

Is there an issue for it? This is a change to how evidence is recorded, so under your methodology it needs one.

Tell me what was agreed and I'll fit your two rows into it. If the answer is "we decided it in a chat and nobody's written it down yet," say so — that's worth knowing too, because a marker reading the repository will see the old structure and no record of the decision.

pull mst recent and ill look

Your register edit is uncommitted and `git pull` may overwrite it. Stash first:

powershell

```
git stash push -m "ai register rows 6-7 august" evidence/ai/ai-usage-register.csv
git pull origin main
```

Then look at what's there:

powershell

```
ls evidence/ai -Recurse | Select-Object -ExpandProperty FullName
```

Paste that. If the per-member structure has landed, your stashed rows go into your own file instead and I'll adjust them. If it hasn't, restore with `git stash pop` and we carry on as before.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git stash push -m "ai register rows
6-7 august" evidence/ai/ai-usage-register.csv
>> git pull origin main
No local changes to save
remote: Failed to authenticate user
fatal: Authentication failed for 'https://sdp.ms.wits.ac.za/git-push-pray/Sport-
Analytics-Tool.git/'
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Two things.

"No local changes to save" — your register edit is gone. You ran `git checkout --` on the file earlier as the first step of the fix, and evidently didn't get as far as re-appending. No loss; I still have the rows and can regenerate them once we know the format.

Authentication failed — your handover says Gitea auth fails intermittently and has always cleared on retry:

```
powershell
```

```
git pull origin main
```

Run it again. If it fails a second time, run it a third; if it's still failing after that, it's not the usual flakiness and worth checking whether anyone else can reach the server.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git pull origin main remote: Enumerating objects: 122, done. remote: Counting objects: 100% (122/122), done...
```

PASTED

Substantial pull. Relevant bits:

The per-member register folder has not landed. `evidence/ai/ai-usage-register.csv` was modified (27 lines changed), not replaced. So the split you described is agreed but not implemented, and the single file is still the register. Your two rows go in it as planned.

Two things that answer earlier concerns: `evidence/sprints/sprint-1/2026-08-06-planning.md` exists, so sprint planning records are being kept. And Shayna and Dean now have real transcripts — the "four members with an empty folder" problem is down to two, Gabriel and Nadav.

The project appears to have a name now — "Stats the Game", with brand guidelines and logos. Nobody told you, which is worth noting.

Check what changed in the register before appending, since 27 lines moved and your rows must not collide:

```
powershell
```

```
git log -1 --format="%an %s" -- evidence/ai/ai-usage-register.csv
Get-Content evidence/ai/ai-usage-register.csv | Select-Object -Last 2
```

Paste both. Then I'll give you the append command with the trailing-newline fix, and you still owe me an issue number.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git log -1 --format="%an %s" --
evidence/ai/ai-usage-register.csv
>> Get-Content evidence/ai/ai-usage-register.csv | Select-Object -Last 2
Nadav Sundy Merge remote-tracking branch 'origin/main' into docs/82-add-brand-
guidelines
"2026-08-07","Shayna Underslak","ChatGPT Web","GPT-5.6 Sol","Planning; backlog
design; documentation; review; Git and Pull Request guidance","Issue #36: establish
and prioritise the complete project backlog; map Sprint 1 work to existing Gitea
```

issues; document the complete Sprint 2, Sprint 3 and Final Submission backlog; define progressive future-sprint refinement; prepare Sprint 1 planning evidence";docs/planning/project-backlog.md; evidence/sprints/sprint-1/2026-08-06-planning.md; MkDocs navigation update; complete Sprint 2, Sprint 3 and Final Submission roadmap; Pull Request description";"Reviewed and repeatedly adapted the backlog structure and issue mappings before use; corrected the planning approach so the complete future backlog is documented during Sprint 1 while detailed future Gitea issues are created progressively; formatted all modified planning files; git diff --check passed; strict MkDocs build passed; repository checks were completed; generated site output and unrelated files were excluded from the Pull Request";"evidence/ai/transcripts/shayna-underslak/2026-08-07-issues-73-36-architecture-backlog.md; Issue #36; linked Pull Request"2026-08-07,Nadav Sundy,ChatGPT Web,GPT-5.6 Sol,Branding; documentation; design review,"Issue #82: define the Stat'sTheGame visual identity, Day Match and Night Match themes, logo variants, asset locations, favicon safe area, accessibility rules and frontend design guidelines.",Brand guidelines document; approved logo asset variants; padded favicon; asset-placement guidance; MkDocs and Git workflow guidance,"Logo concept and both themes were reviewed and approved; guidelines were checked against the current project architecture, repository structure, methodology and AI policy. MkDocs strict build, repository structure check, lint, typecheck, automated tests, production builds and git diff check passed before commit.",Issue #82 and linked Pull Request
2026-08-07,Nadav Sundy,ChatGPT Web,GPT-5.6 Sol,Branding; documentation; design review,"Issue #82: define the Stat'sTheGame visual identity, Day Match and Night Match themes, logo variants, asset locations, favicon safe area, accessibility rules and frontend design guidelines.",Brand guidelines document; approved logo asset variants; padded favicon; asset-placement guidance; MkDocs and Git workflow guidance,"Logo concept and both themes were reviewed and approved; guidelines were checked against the current project architecture, repository structure, methodology and AI policy. MkDocs strict build, repository structure check, lint, typecheck, automated tests, production builds and git diff check passed before commit.",Issue #82 and linked Pull Request
PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more

The register is broken in `main`, and worse than my near-miss earlier.

Three defects:

1. **Nadav's row is fused onto the end of Shayna's** — `...linked Pull Request"2026-08-07,Nadav Sundy,...`. Exactly the missing-trailing-newline bug I hit. It's now committed.
2. **His row is duplicated** — the identical entry appears twice, once fused and once on its own line.
3. **His rows are unquoted** while everything around them is quoted, and they contain commas inside fields. Same defect Liora had, reintroduced.

So the file doesn't parse, and the fix isn't yours to make silently — it's someone else's committed work in `main`.

What I'd do, in order:

Tell Nadav now, not Tuesday. He's the last committer and it's his row; he can fix it in minutes and it's less awkward than you editing his entry.

Then append yours once his fix lands, so you're not layering onto a broken file and inheriting the blame for it.

There's also a pattern worth naming at Thursday's standup: this is the third time the register has been corrupted — Liora, then very nearly you, now Nadav twice over. The per-member split you mentioned would eliminate the class of problem entirely. That's a reason to actually schedule it rather than leave it as a chat decision, and it needs an issue.

Want me to draft the message to Nadav?

No we decided we making files, Ill make mine first, it must have all of mine

Then your file needs every entry you've made, not just the two new ones. That means extracting your existing rows from the shared register.

Step 1 — see exactly what's there under your name:

```
powershell
```

```
Select-String -Path evidence\ai\ai-usage-register.csv -Pattern "Ben Swartz"
```

Paste the output. I recall one entry, 5 August for issue #13, but I want the exact text rather than my reconstruction of it.

Two questions before I write the file:

Does the shared register keep your rows or lose them? If everyone's rows move out, the shared file becomes a header and nothing else — which is fine, but someone has to decide whether it's deleted, kept as a combining target, or replaced by a README pointing at the per-member files. If you remove yours and nobody else removes theirs, the register is half-migrated and worse than either state.

Where do the files go and what are they called? `evidence/ai/registers/ben-swartz.csv` is the obvious guess given `transcripts/ben-swartz/` already exists, but if the group agreed something else, use theirs.

Answer those and paste the grep, and I'll give you the file.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Select-String -Path evidence\ai\ai-usage-register.csv -Pattern "Ben Swartz"
evidence\ai\ai-usage-register.csv:7:"2026-08-05","Ben Swartz","Claude Web","Claude Opus
5","Planning; code generation; technical guidance; documentation","Issue #13: provision the
```

hosted PostgreSQL instance and establish verified backend access"; "Database connection check script; issue and Pull Request descriptions; connection-variant and TLS troubleshooting guidance"; "Ran the connection check successfully against the hosted database; verified the connection variant by DNS and TCP testing rather than accepting the default; kept TLS certificate verification enabled rather than disabling it to pass"; "evidence/ai/transcripts/ben-swartz/ (historical transcript import pending); Issue #13 and linked Pull Request"

PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more

One existing row. So your file holds three: 5, 6 and 7 August.

I'm going with `evidence/ai/registers/ben-swartz.csv`, mirroring `transcripts/ben-swartz/`. Say if the group agreed something else.

Create the file:

powershell

```
New-Item -ItemType Directory -Force -Path evidence\ai\registers | Out-Null
```

powershell

@'

```
"Date","Team member","Tool","Model","Purpose","Brief task","Output used","Verification  
"2026-08-05","Ben Swartz","Claude Web","Claude Opus 5","Planning; code generation; tec  
"2026-08-06","Ben Swartz","Claude Web","Claude Opus 5","Planning; technical guidance;  
"2026-08-07","Ben Swartz","Claude Web","Claude Opus 5","Technical guidance; code gener  
'@ | Set-Content -Path evidence\ai\registers\ben-swartz.csv -Encoding utf8
```

I've replaced "(historical transcript import pending)" in the 5 August row with the actual PDF path, since the transcript now exists.

Verify four lines:

powershell

```
(Get-Content evidence\ai\registers\ben-swartz.csv | Measure-Object).Count
```

Then two decisions you still owe:

Remove your row from the shared register, or leave it? Duplication across both files is worse than either alone. My view: remove it in the same commit, so the migration is coherent for your part.

The issue number. Still outstanding, and this is now a structural change to how evidence is recorded — it needs one more than the append did.

keep in old register for now, and then i havent opened in issue, write it for me

Keeping both is defensible short-term, but the duplication has to be time-boxed or it becomes the permanent state. The issue below says so explicitly.

Issue title:

Establish per-member AI usage registers

Issue body, in your template format:

markdown

Description

Replace the single shared `evidence/ai/ai-usage-register.csv` with one register file per team member under `evidence/ai/registers/`, named to match the existing transcript folders.

The shared register remains in place during the migration. Entries will exist in both files until every member has created their own, after which the shared register is removed and any remaining entries transferred.

Motivation

The shared register is a single append-only file edited by all six members, and it conflicts on almost every merge. It has been corrupted three times: two rows were committed unquoted while containing commas, one row was appended without a preceding newline and fused onto the row above it, and one row was committed twice. The file does not currently parse as CSV.

The AI policy requires attribution for every document and every AI-assisted commit, and states that a document without a declaration will not be marked. A register that does not parse is weak evidence of compliance.

Splitting the file by member removes the conflict entirely, because no two people write to the same file.

Acceptance Criteria

- [] `evidence/ai/registers/` exists and contains one CSV per team member
- [] Each file carries the same nine-column header as the shared register
- [] Every field in every row is quoted
- [] Every file ends with a trailing newline
- [] Each member's existing entries are transferred to their own file
- [] `evidence/ai/README.md` documents the structure, the column meanings, and the requirement that fields are quoted
- [] The shared register is removed once all six files are complete, with a date agreed for this
- [] Every file parses as CSV

Dependencies

Each member must create their own file. This issue cannot be completed by one person alone.

Evidence

Link the Pull Requests creating each member's register.

Two things I'd flag when you post it: assign it to yourself for the structure and your own file, but the other five files need owners or this stalls at one-sixth done. And put a date on removing the shared register — without one, "for now" becomes indefinite and you'll have two sources of truth at Sprint 1.

Once you have the number, tell me and I'll give you the branch and commit commands.

why type is it?

`type: documentation`.

It's a change to how evidence is recorded, not application behaviour. `type: infrastructure` would be defensible if you read the register as project tooling, but your methodology lists documentation as covering exactly this kind of work, and the register is documentation of AI usage.

Area label: `area: documentation`.

Also worth adding `priority: high` — the AI policy requirement is hard (no declaration, not marked), the file currently doesn't parse, and Sprint 1 is eighteen days out.

Bug
Record incorrect behaviour with reproducible evidence
Feature
Implement a user-visible or platform capability
Task
Record documentation, infrastructure, research or general project work
Default
Create an issue from default template.
only have this

Task — its description explicitly covers documentation and general project work.

Bug is tempting since the register is genuinely broken, but Bug is for incorrect behaviour in the product. This is a change to how you record evidence, and the corruption is the motivation rather than the thing being fixed.

Fill in the template's fields with the body I gave you, adapting section headings to whatever the Task template asks for. Labels: `type: documentation`, `area: documentation`, `priority: high`.

opened now by [BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) · 0 comments
[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) commented [now]
(https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues/84#issue-158)

Member

Description

Replace the single shared `evidence/ai/ai-usage-register.csv` with one register

file per team member under `evidence/ai/registers/`, named to match the existing

transcript folders.

The shared register remains in place during the migration. Entries will exist in both files until every member has created their own, after which the shared register is removed and any remaining entries transferred.

Motivation

The shared register is a single append-only file edited by all six members, and it conflicts on almost every merge. It has been corrupted three times: two rows were committed unquoted while containing commas, one row was appended without a

preceding newline and fused onto the row above it, and one row was committed twice. The file does not currently parse as CSV.

The AI policy requires attribution for every document and every AI-assisted commit, and states that a document without a declaration will not be marked. A register that does not parse is weak evidence of compliance.

Splitting the file by member removes the conflict entirely, because no two people write to the same file.

Acceptance Criteria

- `evidence/ai/registers/` exists and contains one CSV per team member
- Each file carries the same nine-column header as the shared register
- Every field in every row is quoted
- Every file ends with a trailing newline
- Each member's existing entries are transferred to their own file
- `evidence/ai/README.md` documents the structure, the column meanings, and the requirement that fields are quoted
- The shared register is removed once all six files are complete, with a date agreed for this
- Every file parses as CSV

Dependencies

Each member must create their own file. This issue cannot be completed by one person alone.

Evidence

Link the Pull Requests creating each member's register.

[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) added the
[area: documentation](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues?labels=9)

label now

[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) self-assigned this now

WritePreview

Drop files or click here to upload.

[area: documentation](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues?labels=9)

No Milestone

No project

[BenSwartz](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues/84#)

1 Participants

Notifications
Due Date
No due date set.
Dependencies
No dependencies set.
Reference: git-push-pray/Sport-Analytics-Tool#84
[Powered by Gitea](https://about.gitea.com/)Version: 1.24.7
Page:85msTemplate:16ms
Show more

#84 is open, but three fields are missing that your methodology requires before an issue can move to **Ready**:

- **No milestone.** Set it to Sprint 1.
- **No project.** It needs to be on the board, or it's invisible to the tracking your Sprint 1 mark depends on.
- **Only one label.** Add **type: documentation** and **priority: high**.

Set those, then:

powershell

```
git switch -c docs/84-add-per-member-ai-register
```

powershell

```
git add evidence/ai/registers/ben-swartz.csv apps/backend/scripts/queries/d2-storage.s
git commit -m "docs(evidence): add a per-member AI usage register" -m "The shared regi
```

The shared register keeps my entries until the migration completes.

Refs #84

Assisted-by: Claude-Web[Claude Opus 5]"

powershell

```
git push -u origin docs/84-add-per-member-ai-register
```

d2-storage.sql is in there because it's been untracked since the storage measurement and belongs in the repository. If you'd rather it went in separately, drop it from the **git add** and we'll do it on its own branch.

Paste the push output, then open the PR.

Stray keystroke again. Run the commands from my last message when you're ready — set the milestone, project and labels on #84 first, then the branch, commit and push.

Establish per-member AI usage registers #84

Open

opened now by [BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) · 0 comments

[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) commented [now]

(https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues/84#issue-158)

Member

Description

Replace the single shared `evidence/ai/ai-usage-register.csv` with one register

file per team member under `evidence/ai/registers/`, named to match the existing transcript folders.

The shared register remains in place during the migration. Entries will exist in both files until every member has created their own, after which the shared register is removed and any remaining entries transferred.

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preceding newline and fused onto the row above it, and one row was committed twice. The file does not currently parse as CSV.

The AI policy requires attribution for every document and every AI-assisted commit, and states that a document without a declaration will not be marked. A register that does not parse is weak evidence of compliance.

Splitting the file by member removes the conflict entirely, because no two people write to the same file.

Acceptance Criteria

- `evidence/ai/registers/` exists and contains one CSV per team member
- Each file carries the same nine-column header as the shared register
- Every field in every row is quoted
- Every file ends with a trailing newline
- Each member's existing entries are transferred to their own file
- `evidence/ai/README.md` documents the structure, the column meanings, and the requirement that fields are quoted
- The shared register is removed once all six files are complete, with a date agreed for this
- Every file parses as CSV

Dependencies

Each member must create their own file. This issue cannot be completed by one person alone.

Evidence

Link the Pull Requests creating each member's register.

[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) added the

[area: documentation](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues?labels=9)

label now

[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) self-assigned this now
[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) added this to the Sprint 1 milestone now
[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) added this to the Sport-Analytics-Tool-Proj project now
[BenSwartz](https://sdp.ms.wits.ac.za/BenSwartz) added the
[priority: high](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues?labels=37)
label now
WritePreview
Drop files or click here to upload.
[area: documentation](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues?labels=9)
[priority: high](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues?labels=37)
[Sprint 1](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/milestone/1)
[Sport-Analytics-Tool-Proj](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/projects/2)
[BenSwartz](https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool/issues/84#)
1 Participants
Notifications
Due Date
No due date set.
Dependencies
No dependencies set.
Reference: git-push-pray/Sport-Analytics-Tool#84
[Powered by Gitea](https://about.gitea.com/)Version: 1.24.7
Page:89msTemplate:18ms

Show more

Milestone, project and priority all set. `type: documentation` still isn't on there — add it if the label exists; if your repository only has the three issue templates and no `type:` labels, then `area: documentation` carries it and that's fine.

Branch and commit:

```
powershell
```

```
git switch -c docs/84-add-per-member-ai-register
```

```
powershell
```

```
git add evidence/ai/registers/ben-swartz.csv apps/backend/scripts/queries/d2-storage.s
```

```
powershell
```

```
git commit -m "docs(evidence): add a per-member AI usage register" -m "The shared regi
```

The shared register keeps my entries until the migration completes.

Refs #84

Assisted-by: Claude-Web[Claude Opus 5]"

powershell

```
git push -u origin docs/84-add-per-member-ai-register
```

Paste the output.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git switch -c docs/84-add-per-
member-ai-register
Switched to a new branch 'docs/84-add-per-member-ai-register'
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git add evidence/ai/registers/ben-
swartz.csv apps/backend/scripts/queries/d2-storage.sql
warning: in the working copy of 'apps/backend/scripts/queries/d2-storage.sql', CRLF
will be replaced by LF the next time Git touches it
warning: in the working copy of 'evidence/ai/registers/ben-swartz.csv', CRLF will be
replaced by LF the next time Git touches it
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git commit -m "docs(evidence): add a
per-member AI usage register" -m "The shared register is edited by all six members
and conflicts on almost every merge. It has been corrupted three times and does not
currently parse. This adds one register per member, starting with mine, with every
field quoted and a trailing newline.
>>
>> The shared register keeps my entries until the migration completes.
>>
>> Refs #84
>> Assisted-by: Claude-Web[Claude Opus 5]"
[docs/84-add-per-member-ai-register 31a893f] docs(evidence): add a per-member
AI usage register
2 files changed, 15 insertions(+)
create mode 100644 apps/backend/scripts/queries/d2-storage.sql
create mode 100644 evidence/ai/registers/ben-swartz.csv
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git push -u origin docs/84-add-per-
member-ai-register
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 2.31 KiB | 592.00 KiB/s, done.
Total 11 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a new pull request for 'docs/84-add-per-member-ai-register':
remote: https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-
Tool/pulls/new/docs/84-add-per-member-ai-register
```

remote:
remote: . Processing 1 references
remote: Processed 1 references in total
To <https://sdp.ms.wits.ac.za/git-push-pray/Sport-Analytics-Tool.git>
* [new branch] docs/84-add-per-member-ai-register -> docs/84-add-per-member-ai-register
branch 'docs/84-add-per-member-ai-register' set up to track 'origin/docs/84-add-per-member-ai-register'.
PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more

Pushed. Open the PR at the URL Gitea printed, with this description:

markdown

Purpose

The shared AI usage register is a single append-only file edited by all six members and conflicts on almost every merge. It has been corrupted three times: rows committed unquoted while containing commas, a row appended without a preceding newline and fused onto the row above it, and a row committed twice. The file does not currently parse as CSV.

This begins the migration to one register per member, which removes the conflict entirely because no two people write to the same file.

Changes

- `evidence/ai/registers/ben-swartz.csv`, carrying all three of my entries for 5, 6 and 7 August.
- `apps/backend/scripts/queries/d2-storage.sql`, the storage measurement query from #27, which had been left untracked.

Testing

The file carries the same nine-column header as the shared register, every field is quoted, and it ends with a trailing newline.

Known Limitations

My entries remain in the shared register as well, so they are duplicated until the other five members create their files. The shared register cannot be removed until then, and no date has been agreed for that. Five members still need to create theirs; the issue cannot be completed by one person.

`evidence/ai/README.md` has not been updated to document the new structure. That acceptance criterion remains open.

Related Issue

Refs #84

Note **Refs**, not **Closes** — #84 needs five more files and a README before it's done, and closing it on your PR would leave the migration half-finished with the issue marked complete. Then move #84 to **In Review** on the board and request a review.

What's the title?

Your git methodology says the PR title must contain the issue number, in the form **#12**

Validate event submissions :

#84 Add a per-member AI usage register

lets just make it for them and for the README and then commit and push so its in one PR

Making the README, yes. Making the other five members' register files, no — and I'd push back on that.

Two reasons. Their existing entries live in the shared register and only they can verify the entries are accurate; you'd be transcribing attributions of AI usage on behalf of five people, which is the one kind of content in this project where accuracy is a policy matter and the person named is responsible for it. Second, the issue explicitly says each member must create their own, and doing it for them removes the evidence that they participated — which the peer review multiplier and the group mark both care about.

What I would do instead: create the **directory structure with empty files carrying only the header**, or a README that names the expected filenames. That unblocks everyone without putting words in their mouths, and it takes one commit.

So: README plus five header-only files, and each member fills their own. Say if that works and I'll write both.

If you genuinely want to transcribe their existing rows for them, say so explicitly and I'll do it — but the PR description should then say you did it and they need to verify their own entries before merge.

yes empty files but so they exists

powershell

```
$header = '"Date","Team member","Tool","Model","Purpose","Brief task","Output used","V
foreach ($m in 'dean-feldman','gabriel-raz','liora-rosenberg','nadav-sundy','shayna-un
    Set-Content -Path "evidence\ai\registers\$m.csv" -Value $header -Encoding utf8
}
ls evidence\ai\registers | Select-Object -ExpandProperty Name
```

Then the README:

powershell

@'

AI usage registers

Each team member records their AI usage in their own file under `registers/`. The file is named to match the member's transcript folder under `transcripts/`.

The single shared register at `ai-usage-register.csv` is being retired. It was edited by all six members, conflicted on almost every merge, and was corrupted three times. Entries remain in both files until every member has created their own, after which the shared register is removed.

Columns

Column	Content
Date	The date the work was done, as `YYYY-MM-DD`.
Team member	The member who used the tool.
Tool	The tool used, for example `Claude Web` or `ChatGPT Web`.
Model	The model used, for example `Claude Opus 5` or `GPT-5.6 Thinking`.
Purpose	The kinds of use, separated by semicolons.
Brief task	What the work was, including the issue number where applicable.
Output used	What was produced and actually used.
Verification or adaptation	How the output was checked, tested or changed before use.
Related evidence	Transcript paths, issue numbers and Pull Requests.

Rules

Every field must be quoted, including the date. Fields contain commas and semicolons, and an unquoted field containing a comma breaks the file for every reader.

Every file must end with a trailing newline. A row appended to a file without one is fused onto the row above it.

Use a semicolon, not a comma, to separate items within a field.

Only tools and models genuinely used may be listed. The register is a record of what happened, not a declaration of what was permitted.

Relationship to other attribution

This register does not replace the attribution the course AI policy requires elsewhere. Commits containing AI-generated code carry an `Assisted-by` line, documents carry an AI declaration, and the repository README declares the tools in use.

AI Declaration

The preceding document was generated with the assistance of Claude-Web[Claude Opus 5].
'@ | Set-Content -Path evidence\ai\registers\README.md -Encoding utf8

Note the doubled apostrophe in `member ' 's` — that's PowerShell escaping inside a single-quoted here-string, and it writes out as one apostrophe.

Run both, paste the file listing, then I'll give you the commit.

give me the command

powershell

```
git add evidence/ai/registers
```

powershell

```
git commit -m "docs(evidence): add per-member register files and documentation" -m "Ad
```

Refs #84

```
Assisted-by: Claude-Web[Claude Opus 5]"
```

powershell

```
git push
```

it will go into the PR now hey?

Yes — a PR tracks the branch, so any commit you push to `docs/84-add-per-member-ai-register` appears in the open PR automatically. No need to reopen or recreate it.

If you already opened the PR with the description I gave you, two lines in it are now stale and should be edited:

- Changes should mention the five header-only files and the README.
- Known Limitations says the README hasn't been updated — that criterion is now met, so remove it.

The duplication caveat stays, and `Refs #84` rather than `Closes #84` still stands: five members' entries are still unwritten.

give me the new decsription to put in

markdown

Purpose

The shared AI usage register is a single append-only file edited by all six members and conflicts on almost every merge. It has been corrupted three times: rows committed unquoted while containing commas, a row appended without a preceding newline and fused onto the row above it, and a row committed twice. The file does not currently parse as CSV.

This begins the migration to one register per member, which removes the conflict entirely because no two people write to the same file.

Changes

- `evidence/ai/registers/ben-swartz.csv`, carrying all three of my entries for 5, 6 and 7 August.
- A header-only register file for each of the other five members, so the structure exists and nobody has to create it themselves.
- `evidence/ai/registers/README.md`, documenting the columns, the quoting requirement, the trailing-newline requirement, and the relationship between this register and the attribution the AI policy requires in commits, documents and the repository README.
- `apps/backend/scripts/queries/d2-storage.sql`, the storage measurement query from #27, which had been left untracked.

Testing

Every file carries the same nine-column header, every field in my entries is quoted, and every file ends with a trailing newline.

Known Limitations

Only my register is populated. The other five files carry a header and nothing else, and each member must transfer their own entries – nobody should be writing attributions of AI usage on another member's behalf.

My entries remain in the shared register as well, so they are duplicated until the migration completes. The shared register cannot be removed until all six files are populated, and no date has been agreed for that.

The shared register does not currently parse. That is not fixed here.

Related Issue

Refs #84

good place to stop hey?

Reasonable place, yes. #27 is merged and closed, and #84 is open with a PR needing review.

Two things before you go, both small:

Request the review on #84 now, even if nobody acts until Monday. An unreviewed PR sitting over the weekend is dead time you don't have with Sprint 1 on 25 August.

Tell Nadav the register is broken. His rows are unquoted, duplicated and fused, and it's in `main`. Fixing it is a few minutes for him and awkward for you to do on his behalf.

Still unrecorded, for whenever you pick this up: the four follow-up issues from #27. The super-over exclusion from batting and bowling aggregates is the one that matters — it exists only in this conversation, and if someone writes derivation before it's an issue, career figures go wrong across 99 matches and nobody will notice until validation.

Also worth carrying forward: password reset and account deletion are named requirements with no visible owner, and the Supabase Auth question has never been put to Brendan. Both are Tuesday's meeting, not yours alone.

Good session's work — the ball-number correction in particular was a real defect that would have been very hard to spot later.

DO i just need to do the pdf?
i

The transcript PDF, yes — it's stale. You exported it covering 05 to 07, but today's session has continued well past that export, and all three of your register rows cite it as evidence.

Re-export this conversation and replace `evidence/ai/transcripts/ben-swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf`. Same filename keeps the register citations valid.

That's a separate commit on the same branch:

```
powershell
```

```
git add evidence/ai/transcripts/ben-swartz/  
git commit -m "docs(evidence): update the AI transcript through 7 August" -m "Refs #84"  
git push
```

Beyond that, no — everything else on my list is either awaiting other people (the review, Nadav's fix) or is new work for next session (the four follow-up issues).

its a new transcript so can't we say the 7th-9t?

Two things worth checking there.

The date. Today is 9 August, not 7. Your terminal has been showing 07/08/2026 throughout — but a lot of what we did today was described as "7 August" in commits and register rows on that basis. If today is actually the 9th, then the 7 August register row, the ADR-003 measurement date, and several commit messages all carry the wrong date. Which is right?

The naming. If this is a genuinely new session rather than a continuation of the export you already have, then a separate file is correct — the existing PDF stays as-is and covers what it covers, and the new one covers this session. Your register rows would then need to cite the right file per row: the 5 and 6 August rows point at the old PDF, today's row points at the new one.

Tell me the actual date and whether the existing PDF ends where this session began, and I'll give you the corrected filename and the register fix.

its the 9th today

Then several things need correcting, and one is already in `main`.

What's wrong:

- Today's register row is dated `2026-08-07`, should be `2026-08-09`.
- ADR-003 says the measurement was taken "on 7 August 2026" — that's committed and merged.
- The issue #27 comment is headed "Progress, 7 August".
- The commit for the transcript I just gave you says "through 7 August".

Fix the register row first, since it's on your open branch:

```
powershell
(Get-Content evidence\ai\registers\ben-swartz.csv) -replace '^"2026-08-07"', '"2026-08-
```

Then check it:

```
powershell
Get-Content evidence\ai\registers\ben-swartz.csv | Select-Object -Last 1
```

The new transcript should then be `2026-08-09-continuous-project-session.pdf`, and that row's evidence field needs to point at it rather than the old PDF. Tell me when the register row is fixed and I'll give you the evidence-field correction in the same command.

ADR-003 is the awkward one — it's merged, so correcting it needs its own issue and branch. Small, but a date error in a decision record is exactly the sort of thing that undermines the rest of the document. Worth a `fix(docs)` on a short branch rather than leaving it.

Was 6 August actually the 6th, or is that wrong too?

no no up until a few messages ago it was the seventh, then it was my sabbath, now its the 9th

That changes things — everything dated 7 August is correct, because it was the 7th when you did it. Nothing needs fixing. ADR-003, the issue comment, the register row, the commits: all accurate.

So today is a new session on the 9th, and the only thing to sort is the transcript.

What the 7 August row should cite: the work happened on the 7th and the transcript for it is the existing PDF. If that export genuinely covers through the end of the 7th, the row is already right and you change nothing.

Today's session is separate. If you do register-worthy work today, it gets its own row dated `2026-08-09` citing a new transcript, `2026-08-09-continuous-project-session.pdf`. So far today you've created #84, made the register files and the README, and opened a PR — that's register-worthy.

So: one new row, one new transcript file. Want me to write the row?

yes

```
powershell
```

```
@'
```

```
"2026-08-09","Ben Swartz","Claude Web","Claude Opus 5","Planning; documentation; issue  
'@ | Add-Content -Path evidence\ai\registers\ben-swartz.csv -Encoding utf8
```

Check it landed on its own line:

```
powershell
```

```
Get-Content evidence\ai\registers\ben-swartz.csv | Select-Object -Last 2
```

Then export today's conversation as `2026-08-09-continuous-project-session.pdf` into `evidence/ai/transcripts/ben-swartz/`, and commit both together:

```
powershell
```

```
git add evidence/ai/registers/ben-swartz.csv evidence/ai/transcripts/ben-swartz/  
git commit -m "docs(evidence): register AI usage for 9 August" -m "Refs #84  
Assisted-by: Claude-Web[Claude Opus 5]"  
git push
```

but the register covers things from the 7th?

DO i just need to do the pdf? i

that was the first message on the 9th

You're right, and it's worth being precise about where the boundary falls.

Everything from the start of this conversation through the PR description for #84 happened on the 7th — C3, C4, the ball-number fix, ADR-003, the merge, creating #84, the register files, the README. Only from "DO i just need to do the pdf?" onward is the 9th.

So the 9 August work is: noticing the transcript was stale, sorting out the dating, and this register row. That's thin, and the row I wrote overstates it badly — it claims #84's drafting and the register structure, which were the 7th's work.

Two options:

Fold it into the 7 August row. The 9th added nothing substantive; the register entry for the 7th already covers the work, and it just needs its evidence field extended to cite both transcripts. Cleanest, and honest.

Or a minimal 9 August row saying what today actually was — reviewing the session's dating and completing the transcript evidence. Accurate but close to trivial.

I'd take the first. The register records AI usage on project work, and today's usage was housekeeping on the 7th's output.

Either way you still need the 9 August transcript file, since this conversation continued past the last export and the policy allows an unedited transcript to be requested.

Which do you want?

minimla 9th row

```
powershell
```

```
@'
"2026-08-09","Ben Swartz","Claude Web","Claude Opus 5","Review; documentation","Comple
 '@ | Add-Content -Path evidence\ai\registers\ben-swartz.csv -Encoding utf8
```

Verify:

```
powershell
```

```
Get-Content evidence\ai\registers\ben-swartz.csv | Select-Object -Last 2
```

Then export this conversation as `2026-08-09-continuous-project-session.pdf` into `evidence/ai/transcripts/ben-swartz/`, and:

```
powershell
```

```
git add evidence/ai/registers/ben-swartz.csv evidence/ai/transcripts/ben-swartz/
git commit -m "docs(evidence): register AI usage for 9 August" -m "Refs #84"
Assisted-by: Claude-Web[Claude Opus 5]"
git push
```

```
PS C:\Users\BenSwartz\Documents\todo-app> '@'
>> "2026-08-09","Ben Swartz","Claude Web","Claude Opus 5","Review;
documentation","Complete the AI evidence for the 7 August session: confirm the
register entries and dating, and export the outstanding transcript","Register
correction; transcript export","Checked each register row against the date the
work was actually done rather than the date of writing; identified that the existing
transcript export ended before the session did and exported the
remainder","evidence/ai/transcripts/ben-swartz/2026-08-09-continuous-project-
session.pdf; Issue #84"
>> '@ | Add-Content -Path evidence\ai\registers\ben-swartz.csv -Encoding utf8
Add-Content : Could not find a part of the path
C:\Users\BenSwartz\Documents\todo-app\evidence\ai\registers\ben-swartz.csv'.
At line:3 char:6
+ '@ | Add-Content -Path evidence\ai\registers\ben-swartz.csv -Encoding ...
+
+ ~~~~~
+ ~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\BenSwa...\ben-
swartz.csv:String) [Add-Content], DirectoryNotFoun
dException
+ FullyQualifiedErrorId :
GetContentWriterDirectoryNotFoundError,Microsoft.PowerShell.Commands.Add
ContentCommand
PS C:\Users\BenSwartz\Documents\todo-app>
```

Wrong directory — you're in `Documents\todo-app`, not the project.

```
powershell
```

```
cd C:\Users\BenSwartz\Sport-Analytics-Tool
```

Then check you're still on the right branch before appending:

```
powershell  
git branch --show-current
```

Should be `docs/84-add-per-member-ai-register`. Paste that, then re-run the append.

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> git branch --show-current  
docs/84-add-per-member-ai-register  
PS C:\Users\BenSwartz\Sport-Analytics-Tool>
```

Right branch. Re-run the append:

```
powershell  
  
@'  
"2026-08-09","Ben Swartz","Claude Web","Claude Opus 5","Review; documentation","Comple  
'@ | Add-Content -Path evidence\ai\registers\ben-swartz.csv -Encoding utf8
```

Then:

```
powershell  
  
Get-Content evidence\ai\registers\ben-swartz.csv | Select-Object -Last 2
```

```
PS C:\Users\BenSwartz\Sport-Analytics-Tool> Get-Content  
evidence\ai\registers\ben-swartz.csv | Select-Object -Last 2  
"2026-08-07","Ben Swartz","Claude Web","Claude Opus 5","Technical guidance;  
code generation; documentation; review","Issue #27: confirm super-over and  
miscounted-over ingestion, correct the printed ball number, measure storage, and  
complete the Pull Request","Corpus inspection scripts; ad-hoc SQL runner and  
validation queries; ball-number ingestion fix; ADR-003 storage amendment;  
schema documentation updates; issue and Pull Request wording","Verified every  
result by SQL rather than by the ingestion summary; identified that the ball  
number was populated with the array position, contradicting property O1 in  
docs/database/schema.md, and corrected it; the storage measurement was found  
insufficient at four fixtures and recorded as outstanding rather than extrapolated;  
reviewed and merged by another team member","evidence/ai/transcripts/ben-  
swartz/2026-08-05-to-2026-08-07-continuous-project-session.pdf; Issue #27 and  
linked Pull Request"  
"2026-08-09","Ben Swartz","Claude Web","Claude Opus 5","Review;
```

documentation","Complete the AI evidence for the 7 August session: confirm the register entries and dating, and export the outstanding transcript","Register correction; transcript export","Checked each register row against the date the work was actually done rather than the date of writing; identified that the existing transcript export ended before the session did and exported the remainder","evidence/ai/transcripts/ben-swartz/2026-08-09-continuous-project-session.pdf; Issue #84"

PS C:\Users\BenSwartz\Sport-Analytics-Tool>

Show more

Both rows clean, on separate lines, properly quoted.

Now export this conversation as `2026-08-09-continuous-project-session.pdf` into `evidence\ai\transcripts\ben-swartz\`, then:

powershell

```
git add evidence/ai/registers/ben-swartz.csv evidence/ai/transcripts/ben-swartz/
```

powershell

```
git commit -m "docs(evidence): register AI usage for 9 August" -m "Refs #84  
Assisted-by: Claude-Web[Claude Opus 5]"
```

powershell

```
git push
```

That completes the branch. The PR description you already have still stands — the 9 August row and transcript are evidence housekeeping and don't change what the PR does.

    just now

